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Partial Differential Equations, Second Edition

1. Introduction

1. Classify each of the partial differential equations in §1.2 as follows:

(a) Is the PDE linear, semilinear, quasilinear or fully nonlinear?

(b) What is the order of the PDE?

2. Let k be a positive integer. Show that a smooth function defined on $\mathbb{R}^n$ has in general

$$\binom{n+k-1}{k} = \binom{n+k-1}{n-1}$$

distinct partial derivatives of order k.

(Hint: This is the number of ways of inserting $n-1$ dividers $|$ within a row of k symbols $\circ$: for example, $\circ\circ\|\circ\circ\circ|\circ|\circ\circ\circ\|\circ\circ\circ\circ|$. Explain why each such pattern corresponds to precisely one of the partial derivatives of order k.)

3. Prove the Multinomial Theorem:

$$(x_1 + \cdots + x_n)^k = \sum_{|\alpha|=k} \binom{|\alpha|}{\alpha} x^\alpha,$$

where $\binom{|\alpha|}{\alpha} := \frac{|\alpha|!}{\alpha!}$, $\alpha! = \alpha_1!\alpha_2! \cdots \alpha_n!$, and $x^\alpha = x_1^{\alpha_1} \cdots x_n^{\alpha_n}$. The sum is taken over all multiindices $\alpha = (\alpha_1, \ldots, \alpha_n)$ with $|\alpha| = k$.

4. Prove Leibniz's formula:

$$D^\alpha(uv) = \sum_{\beta \le \alpha} \binom{\alpha}{\beta} D^\beta u D^{\alpha-\beta} v,$$

where $u, v : \mathbb{R}^n \to \mathbb{R}$ are smooth, $\binom{\alpha}{\beta} := \frac{\alpha!}{\beta!(\alpha-\beta)!}$, and $\beta \le \alpha$ means $\beta_i \le \alpha_i$ $(i = 1, \ldots, n)$.

5. Assume that $f : \mathbb{R}^n \to \mathbb{R}$ is smooth. Prove

$$f(x) = \sum_{|\alpha| \leq k} \frac{1}{\alpha!} D^\alpha f(0) x^\alpha + O(|x|^{k+1}) \qquad \text{as } x \to 0$$

for each $k = 1, 2, \ldots$. This is Taylor's formula in multiindex notation.

(Hint: Fix $x \in \mathbb{R}^n$ and consider the function of one variable $g(t) := f(tx)$.)

2. Four Important Linear Partial Differential Equations

1. All given functions are assumed smooth, unless otherwise stated.

Write down an explicit formula for a function u solving the initial-value problem

$$\begin{cases} u_t + b \cdot Du + cu = 0 & \text{in } \mathbb{R}^n \times (0, \infty), \\ u = g & \text{on } \mathbb{R}^n \times \{t = 0\}. \end{cases}$$

Here $c \in \mathbb{R}$ and $b \in \mathbb{R}^n$ are constants.

2. All given functions are assumed smooth, unless otherwise stated.

Prove that Laplace's equation $\Delta u = 0$ is rotation invariant; that is, if O is an orthogonal $n \times n$ matrix and we define

$$v(x) := u(Ox) \qquad (x \in \mathbb{R}^n),$$

then $\Delta v = 0$.

3. All given functions are assumed smooth, unless otherwise stated.

Modify the proof of the mean-value formulas to show for $n \geq 3$ that

$$u(0) = -\!\!\!\!\!\!\int_{\partial B(0,r)} g\, dS + \frac{1}{n(n-2)\alpha(n)} \int_{B(0,r)} \left(\frac{1}{|x|^{n-2}} - \frac{1}{r^{n-2}} \right) f\, dx,$$

provided

$$\begin{cases} -\Delta u = f & \text{in } B^0(0,r), \\ u = g & \text{on } \partial B(0,r). \end{cases}$$

4. All given functions are assumed smooth, unless otherwise stated.

Give a direct proof that if $u \in C^2(U) \cap C(\bar{U})$ is harmonic within a bounded open set U, then

$$\max_{\bar{U}} u = \max_{\partial U} u.$$

(Hint: Define $u_\varepsilon := u + \varepsilon|x|^2$ for $\varepsilon > 0$, and show u_ε cannot attain its maximum over $\bar{U}$ at an interior point.)

5. All given functions are assumed smooth, unless otherwise stated.

We say $v \in C^2(\bar{U})$ is subharmonic if

$$-\Delta v \leq 0 \quad \text{in } U.$$

(a) Prove for a subharmonic function v that

$$v(x) \leq \fint_{B(x,r)} v \, dy \quad \text{for all } B(x,r) \subset U.$$

(b) Prove that therefore $\max_{\bar{U}} v = \max_{\partial U} v$ if U is bounded.

(c) Let $\phi : \mathbb{R} \to \mathbb{R}$ be smooth and convex. Assume u is harmonic and $v := \phi(u)$. Prove v is subharmonic.

(d) Prove $v := |Du|^2$ is subharmonic, whenever u is harmonic.

6. All given functions are assumed smooth, unless otherwise stated.

Let U be a bounded, open subset of $\mathbb{R}^n$. Prove that there exists a constant C, depending only on U, such that

$$\max_{\bar{U}} |u| \leq C \left(\max_{\partial U} |g| + \max_{\bar{U}} |f| \right)$$

whenever u is a smooth solution of

$$\begin{cases} -\Delta u = f & \text{in } U, \\ u = g & \text{on } \partial U. \end{cases}$$

(Hint: $-\Delta(u + \frac{|x|^2}{2n}\lambda) \leq 0$, for $\lambda := \max_{\bar{U}} |f|$.)

7. All given functions are assumed smooth, unless otherwise stated.

Use Poisson's formula for the ball to prove

$$r^{n-2}\frac{r-|x|}{(r+|x|)^{n-1}}u(0) \leq u(x) \leq r^{n-2}\frac{r+|x|}{(r-|x|)^{n-1}}u(0)$$

whenever u is positive and harmonic in $B^0(0,r)$. This is an explicit form of Harnack's inequality.

8. All given functions are assumed smooth, unless otherwise stated.

Prove Theorem 15 in §2.2.4. (Hint: Since $u \equiv 1$ solves (44) for $g \equiv 1$, the theory automatically implies

$$\int_{\partial B(0,1)} K(x,y)\, dS(y) = 1$$

for each $x \in B^0(0,1)$.)

9. All given functions are assumed smooth, unless otherwise stated.

Let u be the solution of

$$\begin{cases} \Delta u = 0 & \text{in } \mathbb{R}^n_+, \\ u = g & \text{on } \partial\mathbb{R}^n_+, \end{cases}$$

given by Poisson's formula for the half-space. Assume g is bounded and $g(x) = |x|$ for $x \in \partial\mathbb{R}^n_+$, $|x| \leq 1$. Show Du is not bounded near $x = 0$.

(Hint: Estimate $\frac{u(\lambda e_n) - u(0)}{\lambda}$.)

10. All given functions are assumed smooth, unless otherwise stated.

(Reflection principle)

(a) Let U_+ denote the open half-ball $\{x \in \mathbb{R}^n \mid |x| < 1,\ x_n > 0\}$. Assume $u \in C^2(U_+)$ is harmonic in U_+, with $u = 0$ on $\partial U_+ \cap \{x_n = 0\}$. Set

$$v(x) := \begin{cases} u(x) & \text{if } x_n \geq 0, \\ -u(x_1, \ldots, x_{n-1}, -x_n) & \text{if } x_n < 0, \end{cases}$$

for $x \in U = B^0(0,1)$. Prove $v \in C^2(U)$ and thus v is harmonic within U.

(b) Now assume only that $u \in C^2(U_+) \cap C(\overline{U_+})$. Show that v is harmonic within U. (Hint: Use Poisson's formula for the ball.)

11. All given functions are assumed smooth, unless otherwise stated.

(Kelvin transform for Laplace's equation) The Kelvin transform $\mathcal{K}u = \bar{u}$ of a function $u : \mathbb{R}^n \to \mathbb{R}$ is

$$\bar{u}(x) := u(\bar{x})|\bar{x}|^{n-2} = u(x/|x|^2)|x|^{2-n} \qquad (x \neq 0),$$

where $\bar{x} = x/|x|^2$. Show that if u is harmonic, then so is $\bar{u}$.

(Hint: First show that $D_x\bar{x}(D_x\bar{x})^T = |\bar{x}|^4 I$. The mapping $x \mapsto \bar{x}$ is conformal, meaning angle preserving.)

12. All given functions are assumed smooth, unless otherwise stated.

Suppose u is smooth and solves $u_t - \Delta u = 0$ in $\mathbb{R}^n \times (0, \infty)$.

(a) Show $u_\lambda(x,t) := u(\lambda x, \lambda^2 t)$ also solves the heat equation for each $\lambda \in \mathbb{R}$.

(b) Use (a) to show $v(x,t) := x \cdot Du(x,t) + 2tu_t(x,t)$ solves the heat equation as well.

13. All given functions are assumed smooth, unless otherwise stated.

Assume $n = 1$ and $u(x,t) = v(\frac{x}{\sqrt{t}})$.

(a) Show $u_t = u_{xx}$ if and only if

$$v'' + \frac{z}{2}v' = 0. \qquad (*)$$

Show that the general solution of (*) is

$$v(z) = c\int_0^z e^{-s^2/4}\,ds + d.$$

(b) Differentiate $u(x,t) = v(\frac{x}{\sqrt{t}})$ with respect to x and select the constant c properly, to obtain the fundamental solution Φ for $n = 1$. Explain why this procedure produces the fundamental solution. (Hint: What is the initial condition for u?)

14. All given functions are assumed smooth, unless otherwise stated.

Write down an explicit formula for a solution of

$$\begin{cases} u_t - \Delta u + cu = f & \text{in } \mathbb{R}^n \times (0, \infty), \\ u = g & \text{on } \mathbb{R}^n \times \{t = 0\}, \end{cases}$$

where $c \in \mathbb{R}$.

15. All given functions are assumed smooth, unless otherwise stated.

Given $g : [0, \infty) \to \mathbb{R}$, with $g(0) = 0$, derive the formula

$$u(x,t) = \frac{x}{\sqrt{4\pi}} \int_0^t \frac{1}{(t-s)^{3/2}} e^{-\frac{x^2}{4(t-s)}} g(s)\, ds$$

for a solution of the initial/boundary-value problem

$$\begin{cases} u_t - u_{xx} = 0 & \text{in } \mathbb{R}_+ \times (0, \infty), \\ u = 0 & \text{on } \mathbb{R}_+ \times \{t = 0\}, \\ u = g & \text{on } \{x = 0\} \times [0, \infty). \end{cases}$$

(Hint: Let $v(x,t) := u(x,t) - g(t)$ and extend v to $\{x < 0\}$ by odd reflection.)

16. All given functions are assumed smooth, unless otherwise stated.

Give a direct proof that if U is bounded and $u \in C_1^2(U_T) \cap C(\overline{U_T})$ solves the heat equation, then

$$\max_{\overline{U_T}} u = \max_{\Gamma_T} u.$$

(Hint: Define $u_\varepsilon := u - \varepsilon t$ for $\varepsilon > 0$, and show u_ε cannot attain its maximum over $\overline{U_T}$ at a point in U_T.)

17. All given functions are assumed smooth, unless otherwise stated.

We say $v \in C_1^2(U_T)$ is a subsolution of the heat equation if

$$v_t - \Delta v \leq 0 \quad \text{in } U_T.$$

(a) Prove for a subsolution v that

$$v(x,t) \leq \frac{1}{4r^n} \iint_{E(x,t;r)} v(y,s) \frac{|x-y|^2}{(t-s)^2}\, dy\, ds$$

for all $E(x,t;r) \subset U_T$.

(b) Prove that therefore $\max_{\overline{U_T}} v = \max_{\Gamma_T} v$.

(c) Let $\phi : \mathbb{R} \to \mathbb{R}$ be smooth and convex. Assume u solves the heat equation and $v := \phi(u)$. Prove v is a subsolution.

(d) Prove $v := |Du|^2 + u_t^2$ is a subsolution, whenever u solves the heat equation.

18. All given functions are assumed smooth, unless otherwise stated.

Assume u solves the initial-value problem

$$\begin{cases} u_{tt} - \Delta u = 0 & \text{in } \mathbb{R}^n \times (0, \infty), \\ u = 0, \ u_t = h & \text{on } \mathbb{R}^n \times \{t = 0\}. \end{cases}$$

Show that $v := u_t$ solves

$$\begin{cases} v_{tt} - \Delta v = 0 & \text{in } \mathbb{R}^n \times (0, \infty), \\ v = h, \ v_t = 0 & \text{on } \mathbb{R}^n \times \{t = 0\}. \end{cases}$$

This is Stokes' rule.

19. All given functions are assumed smooth, unless otherwise stated.

(a) Show the general solution of the PDE $u_{xy} = 0$ is

$$u(x, y) = F(x) + G(y)$$

for arbitrary functions F, G.

(b) Using the change of variables $\xi = x + t$, $\eta = x - t$, show $u_{tt} - u_{xx} = 0$ if and only if $u_{\xi\eta} = 0$.

(c) Use (a) and (b) to rederive d'Alembert's formula.

(d) Under what conditions on the initial data g, h is the solution u a right-moving wave? A left-moving wave?

20. All given functions are assumed smooth, unless otherwise stated.

Assume that for some attenuation function $\alpha = \alpha(r)$ and delay function $\beta = \beta(r) \geq 0$, there exist for all profiles ϕ solutions of the wave equation in $(\mathbb{R}^n - \{0\}) \times \mathbb{R}$ having the form

$$u(x,t) = \alpha(r)\phi(t - \beta(r)).$$

Here $r = |x|$ and we assume $\beta(0) = 0$.

Show that this is possible only if $n = 1$ or 3, and compute the form of the functions α, β.

(T. Morley, *SIAM Review* 27 (1985), 69–71)

21. All given functions are assumed smooth, unless otherwise stated.

(a) Assume $\mathbf{E} = (E_1, E_2, E_3)$ and $\mathbf{B} = (B_1, B_2, B_3)$ solve Maxwell's equations

$$\begin{cases} \mathbf{E}_t = \operatorname{curl} \mathbf{B}, \quad \mathbf{B}_t = -\operatorname{curl} \mathbf{E}, \\ \operatorname{div} \mathbf{B} = \operatorname{div} \mathbf{E} = 0. \end{cases}$$

Show $\mathbf{E}_{tt} - \Delta \mathbf{E} = 0$, $\mathbf{B}_{tt} - \Delta \mathbf{B} = 0$.

(b) Assume that $\mathbf{u} = (u_1, u_2, u_3)$ solves the evolution equations of linear elasticity

$$\mathbf{u}_{tt} - \mu \Delta \mathbf{u} - (\lambda + \mu) D(\operatorname{div} \mathbf{u}) = 0 \quad \text{in } \mathbb{R}^3 \times (0, \infty).$$

Show $w := \operatorname{div} \mathbf{u}$ and $\mathbf{w} := \operatorname{curl} \mathbf{u}$ each solve wave equations, but with differing speeds of propagation.

22. All given functions are assumed smooth, unless otherwise stated.

Let u denote the density of particles moving to the right with speed one along the real line and let v denote the density of particles moving to the left with speed one. If at rate $d > 0$ right-moving particles randomly become left-moving, and vice versa, we have the system of PDE

$$\begin{cases} u_t + u_x = d(v - u), \\ v_t - v_x = d(u - v). \end{cases}$$

Show that both $w := u$ and $w := v$ solve the telegraph equation

$$w_{tt} + 2dw_t - w_{xx} = 0.$$

23. All given functions are assumed smooth, unless otherwise stated.

Let S denote the square lying in $\mathbb{R} \times (0, \infty)$ with corners at the points $(0, 1)$, $(1, 2)$, $(0, 3)$, $(-1, 2)$. Define

$$f(x,t) := \begin{cases} -1 & \text{for } (x,t) \in S \cap \{t > x + 2\}, \\ 1 & \text{for } (x,t) \in S \cap \{t < x + 2\}, \\ 0 & \text{otherwise.} \end{cases}$$

Assume u solves

$$\begin{cases} u_{tt} - u_{xx} = f & \text{in } \mathbb{R} \times (0, \infty), \\ u = 0, \ u_t = 0 & \text{on } \mathbb{R} \times \{t = 0\}. \end{cases}$$

Describe the shape of u for times $t > 3$.

(J. G. Kingston, *SIAM Review* 30 (1988), 645–649)

24. All given functions are assumed smooth, unless otherwise stated.

(Equipartition of energy) Let u solve the initial-value problem for the wave equation in one dimension:

$$\begin{cases} u_{tt} - u_{xx} = 0 & \text{in } \mathbb{R} \times (0, \infty), \\ u = g, \ u_t = h & \text{on } \mathbb{R} \times \{t = 0\}. \end{cases}$$

Suppose g, h have compact support. The kinetic energy is $k(t) := \frac{1}{2} \int_{-\infty}^{\infty} u_t^2(x,t)\, dx$ and the potential energy is $p(t) := \frac{1}{2} \int_{-\infty}^{\infty} u_x^2(x,t)\, dx$. Prove

(a) $k(t) + p(t)$ is constant in t,

(b) $k(t) = p(t)$ for all large enough times t.

3. Nonlinear First-Order PDE

1. All given functions are assumed smooth, unless otherwise stated.

Prove

$$u(x,t,a,b) = a \cdot x - tH(a) + b \qquad (a \in \mathbb{R}^n,\ b \in \mathbb{R})$$

is a complete integral of the Hamilton–Jacobi equation $u_t + H(Du) = 0$.

2. All given functions are assumed smooth, unless otherwise stated.

Compute the envelopes of the family of lines

$$x_1 + a^2 x_2 - 2a = 0 \qquad (a \in \mathbb{R})$$

in $\mathbb{R}^2$ and of the family of planes

$$2a_1 x_1 + 2a_2 x_2 - x_3 + a_1^2 + a_2^2 = 0 \qquad (a_1, a_2 \in \mathbb{R})$$

in $\mathbb{R}^3$. Draw pictures illustrating the geometric meaning of the envelopes.

3. All given functions are assumed smooth, unless otherwise stated.

Suppose that the formula $G(x, z, a) = 0$ implicitly defines the function $z = u(x, a)$, where $x, a \in \mathbb{R}^n$. Assume further that we can eliminate the variables a from the identities

$$\begin{cases} G(x, u, a) = 0, \\ G_{x_i}(x, u, a) + G_z(x, u, a)u_{x_i} = 0 \quad (i = 1, \dots, n), \end{cases}$$

to solve for $u = u(x)$.

(a) Find a PDE that u solves if $G = \sum_{i=1}^n a_i x_i^2 + z^3$.

(b) What is the PDE characterizing all spheres in $\mathbb{R}^{n+1}$ with unit radius and center in $\mathbb{R}^n \times \{z = 0\}$?

4. All given functions are assumed smooth, unless otherwise stated.

(a) Write down the characteristic equations for the PDE

$$u_t + b \cdot Du = f \quad \text{in } \mathbb{R}^n \times (0, \infty), \tag{*}$$

where $b \in \mathbb{R}^n$, $f = f(x, t)$.

(b) Use the characteristic ODE to solve (*) subject to the initial condition $u = g$ on $\mathbb{R}^n \times \{t = 0\}$. Make sure your answer agrees with formula (5) in §2.1.2.

5. All given functions are assumed smooth, unless otherwise stated.

Solve using characteristics:

(a) $x_1 u_{x_1} + x_2 u_{x_2} = 2u$, $u(x_1, 1) = g(x_1)$.

(b) $x_1 u_{x_1} + 2x_2 u_{x_2} + u_{x_3} = 3u$, $u(x_1, x_2, 0) = g(x_1, x_2)$.

(c) $u u_{x_1} + u_{x_2} = 1$, $u(x_1, x_1) = \frac{1}{2} x_1$.

6. All given functions are assumed smooth, unless otherwise stated.

Given a smooth vector field $\mathbf{b}$ on $\mathbb{R}^n$, let $\mathbf{x}(s) = \mathbf{x}(s, x, t)$ solve the ODE

$$\begin{cases} \dot{\mathbf{x}} = \mathbf{b}(\mathbf{x}) & (s \in \mathbb{R}), \\ \mathbf{x}(t) = x. \end{cases}$$

(a) Define the Jacobian $J(s, x, t) := \det D_x \mathbf{x}(s, x, t)$ and derive Euler's formula:

$$J_s = (\operatorname{div} \mathbf{b}(\mathbf{x}))J.$$

(b) Show that $u(x, t) := g(\mathbf{x}(0, x, t))J(0, x, t)$ solves

$$\begin{cases} u_t + \operatorname{div}(u\mathbf{b}) = 0 & \text{in } \mathbb{R}^n \times (0, \infty), \\ u = g & \text{on } \mathbb{R}^n \times \{t = 0\}. \end{cases}$$

(Hint: Show $\frac{\partial}{\partial s}(u(\mathbf{x}, s)J) = 0$.)

7. All given functions are assumed smooth, unless otherwise stated.

Verify assertion (36) in §3.2.3, that when Γ is not flat near x^0, the noncharacteristic condition is

$$D_p F(p^0, z^0, x^0) \cdot \boldsymbol{\nu}(x^0) \neq 0.$$

8. All given functions are assumed smooth, unless otherwise stated.

Confirm that the formula $u = g(x - t\mathbf{F}'(u))$ from §3.2.5 provides an implicit solution for the conservation law

$$u_t + \operatorname{div} \mathbf{F}(u) = 0.$$

9. All given functions are assumed smooth, unless otherwise stated.

Consider the problem of minimizing the action $\int_0^t L(\dot{\mathbf{w}}(s), \mathbf{w}(s))\, ds$ over the new admissible class

$$\mathcal{A} := \{\mathbf{w}(\cdot) \in C^2([0,t]; \mathbb{R}^n) \mid \mathbf{w}(t) = x\},$$

where we do not require that $\mathbf{w}(0) = y$.

(a) Show that a minimizer $\mathbf{x}(\cdot) \in \mathcal{A}$ solves the Euler–Lagrange equations

$$-\frac{d}{ds}(D_v L(\dot{\mathbf{x}}(s), \mathbf{x}(s))) + D_x L(\dot{\mathbf{x}}(s), \mathbf{x}(s)) = 0 \quad (0 \le s \le t).$$

(b) Prove that $D_v L(\dot{\mathbf{x}}(0), \mathbf{x}(0)) = 0$.

(c) Suppose now that $\mathbf{x}(\cdot) \in \mathcal{A}$ minimizes the modified action

$$\int_0^t L(\dot{\mathbf{w}}(s), \mathbf{w}(s))\, ds + g(\mathbf{w}(0)).$$

Show that $\mathbf{x}(\cdot)$ solves the usual Euler–Lagrange equations and determine the boundary condition at $s = 0$.

10. All given functions are assumed smooth, unless otherwise stated.

If $H : \mathbb{R}^n \to \mathbb{R}$ is convex, we write $L = H^*$.

(a) Let $H(p) = \frac{1}{r}|p|^r$, for $1 < r < \infty$. Show

$$L(v) = \frac{1}{s}|v|^s, \qquad \text{where } \frac{1}{r} + \frac{1}{s} = 1.$$

(b) Let $H(p) = \frac{1}{2}\sum_{i,j=1}^n a_{ij}p_ip_j + \sum_{i=1}^n b_ip_i$, where $A = ((a_{ij}))$ is a symmetric, positive definite matrix, $b \in \mathbb{R}^n$. Compute $L(v)$.

11. All given functions are assumed smooth, unless otherwise stated.

Let $H : \mathbb{R}^n \to \mathbb{R}$ be convex. We say v belongs to the subdifferential of H at p, written $v \in \partial H(p)$, if

$$H(r) \geq H(p) + v \cdot (r - p) \quad \text{for all } r \in \mathbb{R}^n.$$

Prove $v \in \partial H(p)$ if and only if $p \in \partial L(v)$ if and only if $p \cdot v = H(p) + L(v)$, where $L = H^*$.

12. All given functions are assumed smooth, unless otherwise stated.

Assume $L_1, L_2 : \mathbb{R}^n \to \mathbb{R}$ are convex, smooth and superlinear. Show that

$$\min_{v \in \mathbb{R}^n} (L_1(v) + L_2(v)) = \max_{p \in \mathbb{R}^n} (-H_1(p) - H_2(-p)),$$

where $H_1 = L_1^*$, $H_2 = L_2^*$.

13. All given functions are assumed smooth, unless otherwise stated.

Prove that the Hopf–Lax formula reads

$$\begin{aligned} u(x,t) &= \min_{y\in\mathbb{R}^n}\left\{tL\left(\frac{x-y}{t}\right)+g(y)\right\} \\ &= \min_{y\in B(x,Rt)}\left\{tL\left(\frac{x-y}{t}\right)+g(y)\right\} \end{aligned}$$

for $R = \sup_{\mathbb{R}^n} |DH(Dg)|$, $H = L^*$. (This proves finite propagation speed for a Hamilton–Jacobi PDE with convex Hamiltonian and Lipschitz continuous initial function g.)

14. All given functions are assumed smooth, unless otherwise stated.

Let E be a closed subset of $\mathbb{R}^n$. Show that if the Hopf–Lax formula could be applied to the initial-value problem

$$\begin{cases} u_t + |Du|^2 = 0 & \text{in } \mathbb{R}^n \times (0,\infty), \\ u = \begin{cases} 0 & x \in E, \\ +\infty & x \notin E \end{cases} & \text{on } \mathbb{R}^n \times \{t = 0\}, \end{cases}$$

it would give the solution $u(x,t) = \frac{1}{4t}\operatorname{dist}(x,E)^2$.

15. All given functions are assumed smooth, unless otherwise stated.

Provide all details for the proof of Lemma 4 in §3.3.3.

16. All given functions are assumed smooth, unless otherwise stated.

Assume u^1, u^2 are two solutions of the initial-value problems

$$\begin{cases} u_t^i + H(Du^i) = 0 & \text{in } \mathbb{R}^n \times (0,\infty), \\ u^i = g^i & \text{on } \mathbb{R}^n \times \{t=0\} \end{cases} \quad (i=1,2),$$

given by the Hopf–Lax formula. Prove the L^∞-contraction inequality

$$\sup_{\mathbb{R}^n} |u^1(\cdot,t) - u^2(\cdot,t)| \le \sup_{\mathbb{R}^n} |g^1 - g^2| \qquad (t>0).$$

17. All given functions are assumed smooth, unless otherwise stated.

Show that

$$u(x,t) := \begin{cases} -\frac{2}{3}\left(t + \sqrt{3x + t^2}\right) & \text{if } 4x + t^2 > 0, \\ 0 & \text{if } 4x + t^2 < 0 \end{cases}$$

is an (unbounded) entropy solution of $u_t + (\frac{u^2}{2})_x = 0$.

18. All given functions are assumed smooth, unless otherwise stated.

Assume $u(x+z) - u(x) \leq Ez$ for all $z > 0$. Let $u^\epsilon = \eta_\epsilon * u$, and show $u_x^\epsilon \leq E$.

19. All given functions are assumed smooth, unless otherwise stated.

Assume $F(0) = 0$, u is a continuous integral solution of the conservation law

$$\begin{cases} u_t + F(u)_x = 0 & \text{in } \mathbb{R} \times (0, \infty), \\ u = g & \text{on } \mathbb{R} \times \{t = 0\}, \end{cases}$$

and u has compact support in $\mathbb{R} \times [0, T]$ for each time $T > 0$. Prove

$$\int_{-\infty}^{\infty} u(\cdot, t)\, dx = \int_{-\infty}^{\infty} g\, dx$$

for all $t > 0$.

20. All given functions are assumed smooth, unless otherwise stated.

Compute explicitly the unique entropy solution of

$$\begin{cases} u_t + (\frac{u^2}{2})_x = 0 & \text{in } \mathbb{R} \times (0, \infty), \\ u = g & \text{on } \mathbb{R} \times \{t = 0\}, \end{cases}$$

for

$$g(x) = \begin{cases} 1 & \text{if } x < -1, \\ 0 & \text{if } -1 < x < 0, \\ 2 & \text{if } 0 < x < 1, \\ 0 & \text{if } x > 1. \end{cases}$$

Draw a picture documenting your answer, being sure to illustrate what happens for all times $t > 0$.

4. Other Ways to Represent Solutions

1. All given functions are assumed smooth, unless otherwise stated.

Use separation of variables to find a nontrivial solution u of the PDE

$$u_{x_1}^2 u_{x_1x_1} + 2u_{x_1}u_{x_2}u_{x_1x_2} + u_{x_2}^2 u_{x_2x_2} = 0 \quad \text{in } \mathbb{R}^2.$$

(G. Aronsson, *Manuscripta Math.* 47 (1984), 133–151)

2. All given functions are assumed smooth, unless otherwise stated.

Consider Laplace's equation $\Delta u = 0$ in $\mathbb{R}^2$, taken with the Cauchy data

$$u = 0, \quad \frac{\partial u}{\partial x_2} = \frac{1}{n}\sin(nx_1) \quad \text{on } \{x_2 = 0\}.$$

Employ separation of variables to derive the solution

$$u = \frac{1}{n^2}\sin(nx_1)\sinh(nx_2).$$

What happens to u as $n \to \infty$? Is the Cauchy problem for Laplace's equation well-posed? (This example is due to Hadamard.)

3. All given functions are assumed smooth, unless otherwise stated.

Find explicit formulas for v and σ, so that $u(x,t) := v(x - \sigma t)$ is a traveling wave solution of the nonlinear diffusion equation $u_t - u_{xx} = f(u)$, where

$$f(z) = -2z^3 + 3z^2 - z.$$

Assume $\lim_{s\to\infty} v = 1$, $\lim_{s\to-\infty} v = 0$, $\lim_{s\to\pm\infty} v' = 0$.

(Hint: Multiply the equation $v'' + \sigma v' + f(v) = 0$ by v' and integrate, to determine the value of σ.)

4. All given functions are assumed smooth, unless otherwise stated.

If we look for a radial solution $u(x) = v(r)$ of the nonlinear elliptic equation $-\Delta u = u^p$ in $\mathbb{R}^n$, where $r = |x|$ and $p > 1$, we are led to the nonautonomous ODE

$$v'' + \frac{n-1}{r} v' + v^p = 0. \tag{*}$$

Show that the Emden–Fowler transformation

$$t := \log r, \qquad x(t) := e^{\frac{2t}{p-1}} v(e^t)$$

converts (*) into an autonomous ODE for the new unknown $x = x(t)$.

5. All given functions are assumed smooth, unless otherwise stated.

Find a nonnegative scaling invariant solution having the form $u(x,t) = t^{-\alpha}v(xt^{-\beta})$ for the nonlinear heat equation

$$u_t - \Delta(u^\gamma) = 0,$$

where $\frac{n-2}{n} < \gamma < 1$. Your solution should go to zero algebraically as $|x| \to \infty$.

6. All given functions are assumed smooth, unless otherwise stated.

Find a solution of

$$-\Delta u + u^{\frac{n+2}{n-2}} = 0 \quad \text{in } B(0,1)$$

having the form $u = \alpha(1 - |x|^2)^{-\beta}$ for positive constants α, β. This example shows that a solution of a nonlinear PDE can be finite within a region and yet approach infinity everywhere on its boundary.

7. All given functions are assumed smooth, unless otherwise stated.

Consider the viscous conservation law

$$u_t + F(u)_x - au_{xx} = 0 \quad \text{in } \mathbb{R} \times (0, \infty), \tag{*}$$

where $a > 0$ and F is uniformly convex.

(a) Show u solves (*) if $u(x,t) = v(x - \sigma t)$ and v is defined implicitly by the formula

$$s = \int_c^{v(s)} \frac{a}{F(z) - \sigma z + b}\, dz \qquad (s \in \mathbb{R}),$$

where b and c are constants.

(b) Demonstrate that we can find a traveling wave satisfying $\lim_{s \to -\infty} v(s) = u_l$, $\lim_{s \to \infty} v(s) = u_r$ for $u_l > u_r$, if and only if

$$\sigma = \frac{F(u_l) - F(u_r)}{u_l - u_r}.$$

(c) Let u^ε denote the above traveling wave solution of (*) for $a = \varepsilon$, with $u^\varepsilon(0,0) = \frac{u_l + u_r}{2}$. Compute $\lim_{\varepsilon \to 0} u^\varepsilon$ and explain your answer.

8. All given functions are assumed smooth, unless otherwise stated.

Prove that if u is the solution of problem (23) for Schrödinger's equation in §4.3 given by formula (20), then

$$\|u(\cdot,t)\|_{L^\infty(\mathbb{R}^n)} \leq \frac{1}{(4\pi|t|)^{n/2}}\|g\|_{L^1(\mathbb{R}^n)}$$

for each $t \neq 0$.

9. All given functions are assumed smooth, unless otherwise stated.

Assume that u solves the nonlinear heat equation

$$u_t = \frac{u_{xx}}{u_x^2} \quad \text{in } \mathbb{R} \times (0, \infty)$$

with $u_x > 0$. Let v denote the inverse function to u in the variable x for each time $t > 0$, so that $y = u(x, t)$ if and only if $x = v(y, t)$.

Show that v solves a linear PDE.

10. All given functions are assumed smooth, unless otherwise stated.

Find a function $f : \mathbb{R}^3 \to \mathbb{R}$, $f = f(z, p_1, p_2)$, so that if u is any solution of the rotated wave equation $u_{xt} = 0$, then $w := f(u, u_x, u_t)$ solves Liouville's equation $w_{xt} = e^w$.

(Hint: Show that f must have the form $f(z, p_1, p_2) = a(z) + b(p_1) + c(p_2)$.)

11. All given functions are assumed smooth, unless otherwise stated.

(Lax pairs) Assume that $\{L(t)\}_{t\geq 0}$ is a family of symmetric linear operators on some real Hilbert space H, satisfying the evolution equation

$$\dot{L} = [B, L] = BL - LB,$$

for some collection of operators $\{B(t)\}_{t\geq 0}$. Suppose also that we have a corresponding family of eigenvalues $\{\lambda(t)\}_{t\geq 0}$ and eigenvectors $\{w(t)\}_{t\geq 0}$:

$$L(t)w(t) = \lambda(t)w(t).$$

Assume that L, B, λ and w all depend smoothly upon the time parameter t. Show that $\dot{\lambda} \equiv 0$.

(Hint: Differentiate the identity $Lw = \lambda w$ with respect to t. Calculate $(\dot{\lambda}w, w)$.)

12. All given functions are assumed smooth, unless otherwise stated.

(Continuation) Given a function $u = u(x,t)$, define the linear operators

$$L(t)v := -v_{xx} + uv, \qquad B(t)v = -4v_{xxx} + 6uv_x + 3u_x v.$$

Show that $(\dot{L} - [B,L])v = (u_t + u_{xxx} - 6uu_x)v$. Consequently, if u solves the KdV equation $u_t - 6uu_x + u_{xxx} = 0$, then the eigenvalues of the linear operators $L(t)$ do not change with time.

13. All given functions are assumed smooth, unless otherwise stated.

Show that we can construct an explicit solution of the initial-value problem (17), (19) from Example 1 in §4.5.1, having the form

$$v(z,t) = \frac{1}{\sigma(t)} \frac{1}{(\pi\gamma(t))^{1/2}} e^{-z^2/\gamma(t)} \qquad (z \in \mathbb{R},\ t > 0),$$

the function $\gamma(t)$ to be found. Substitute into the PDE and determine an ODE that γ should satisfy. What is the initial condition for this ODE?

14. All given functions are assumed smooth, unless otherwise stated.

Let u^ϵ and v^ϵ solve the system

$$\begin{cases} u_t^\epsilon + \frac{1}{\epsilon} u_x^\epsilon = \frac{(v^\epsilon)^2 - (u^\epsilon)^2}{\epsilon^2}, \\ v_t^\epsilon - \frac{1}{\epsilon} v_x^\epsilon = \frac{(u^\epsilon)^2 - (v^\epsilon)^2}{\epsilon^2}. \end{cases}$$

Suppose we can write

$$u^\epsilon = u_0 + \epsilon u_1 + \epsilon u_2^\epsilon, \qquad v^\epsilon = v_0 + \epsilon v_1 + \epsilon v_2^\epsilon,$$

where u_0, v_0, u_1, v_1 are smooth, $u_0, v_0 > 0$, and the functions $u_2^\epsilon, v_2^\epsilon$ are bounded, along with their derivatives, uniformly in ϵ.

Show that $u_0 \equiv v_0 =: w$ and w solves the nonlinear heat equation

$$w_t - \frac{1}{4}(\log w)_{xx} = 0.$$

(T. Kurtz, *Trans. AMS* 186 (1973), 259–272)

15. All given functions are assumed smooth, unless otherwise stated.

Justify in the proof of Lemma 2 in §4.5.3 the transformation of the integral of e^{-z^2} over the line Γ to the integral over the real axis.

16. All given functions are assumed smooth, unless otherwise stated.

Utilize Lemma 2 in §4.5.3 to discuss the sense in which u defined by formula (20) in §4.3.1 converges to the initial data g as $t \to 0^+$.

17. All given functions are assumed smooth, unless otherwise stated.

Let $n = 1$ and suppose that u^ε solves the problem

$$\begin{cases} -(a(\frac{x}{\varepsilon})u_x^\varepsilon)_x = f & \text{in } (0,1), \\ u^\varepsilon(0) = u^\varepsilon(1) = 0, \end{cases}$$

where a is a smooth, positive function that is 1-periodic. Assume also that $f \in L^2(0,1)$.

(a) Show that $u^\varepsilon \rightharpoonup u$ weakly in $H_0^1(0,1)$, where u solves

$$\begin{cases} -\bar{a}u_{xx} = f & \text{in } (0,1), \\ u(0) = u(1) = 0, \end{cases}$$

for $\bar{a} := (\int_0^1 a(y)^{-1}\, dy)^{-1}$.

(b) Check that this answer agrees with the conclusions (73), (74) in §4.5.4.

(This problem requires knowledge of energy estimates, Sobolev spaces, etc., from Chapters 5, 6.)

18. All given functions are assumed smooth, unless otherwise stated.

(Variational principles in homogenization) Let $A(y) = ((a_{ij}(y)))$ be symmetric, positive definite and Q-periodic. Recall from §4.5.4 the expression (74) for the corresponding homogenized coefficients $\bar{A} = ((\bar{a}_{ij}))$.

(a) Derive for each $\xi \in \mathbb{R}^n$ the variational formula

$$\xi \cdot \bar{A}\xi = \min_w \left\{ \int_Q Dw \cdot A(y) Dw \, dy \;\middle|\; w = y \cdot \xi + v,\ v\ Q\text{-periodic} \right\}.$$

(Hint: The minimum is attained by $w = y \cdot \xi - \sum_{i=1}^n \xi_i \chi^i$, for the correctors χ^i introduced in §4.5.4.)

(b) Derive also the dual variational formula

$$\eta \cdot \bar{A}^{-1}\eta = \min_{\boldsymbol{\sigma}} \left\{ \int_Q \boldsymbol{\sigma} \cdot A(y)^{-1} \boldsymbol{\sigma} \, dy \;\middle|\; \int_Q \boldsymbol{\sigma} \, dy = \eta,\ \operatorname{div} \boldsymbol{\sigma} = 0,\ \boldsymbol{\sigma}\ Q\text{-periodic} \right\}.$$

(c) Show that therefore

$$\left(\int_Q A(y)^{-1} \, dy \right)^{-1} \leq \bar{A} \leq \int_Q A(y) \, dy.$$

(Remember from §A.1 that for symmetric matrices $R \geq S$ means $R - S$ is nonnegative definite.)

19. All given functions are assumed smooth, unless otherwise stated.

Show that the line $\{t = 0\}$ is characteristic for the heat equation $u_t = u_{xx}$. Show there does not exist an analytic solution u of the heat equation in $\mathbb{R} \times \mathbb{R}$, with $u = \frac{1}{1+x^2}$ on $\{t = 0\}$.

(Hint: Assume there is an analytic solution, compute its coefficients, and show that the resulting power series diverges except at $(0, 0)$. This example is due to Kovalevskaya.)

5. Sobolev Spaces

1. U denotes an open subset of $\mathbb{R}^n$ with smooth boundary ∂U. All given functions are assumed smooth, unless otherwise stated.

Suppose $k \in \{0, 1, \ldots\}$, $0 < \gamma \leq 1$. Prove $C^{k,\gamma}(\bar{U})$ is a Banach space.

2. U denotes an open subset of $\mathbb{R}^n$ with smooth boundary ∂U. All given functions are assumed smooth, unless otherwise stated.

Assume $0 < \beta < \gamma \leq 1$. Prove the interpolation inequality

$$\|u\|_{C^{0,\gamma}(U)} \leq \|u\|_{C^{0,\beta}(U)}^{\frac{1-\gamma}{1-\beta}} \|u\|_{C^{0,1}(U)}^{\frac{\gamma-\beta}{1-\beta}}.$$

3. U denotes an open subset of $\mathbb{R}^n$ with smooth boundary ∂U. All given functions are assumed smooth, unless otherwise stated.

Denote by U the open square $\{x \in \mathbb{R}^2 \mid |x_1| < 1,\ |x_2| < 1\}$. Define

$$u(x) = \begin{cases} 1 - x_1 & \text{if } x_1 > 0,\ |x_2| < x_1, \\ 1 + x_1 & \text{if } x_1 < 0,\ |x_2| < -x_1, \\ 1 - x_2 & \text{if } x_2 > 0,\ |x_1| < x_2, \\ 1 + x_2 & \text{if } x_2 < 0,\ |x_1| < -x_2. \end{cases}$$

For which $1 \leq p \leq \infty$ does u belong to $W^{1,p}(U)$?

4. U denotes an open subset of $\mathbb{R}^n$ with smooth boundary ∂U. All given functions are assumed smooth, unless otherwise stated.

Assume $n = 1$ and $u \in W^{1,p}(0,1)$ for some $1 \leq p < \infty$.

(a) Show that u is equal a.e. to an absolutely continuous function and u' (which exists a.e.) belongs to $L^p(0,1)$.

(b) Prove that if $1 < p < \infty$, then

$$|u(x) - u(y)| \leq |x - y|^{1-1/p} \left(\int_0^1 |u'|^p \, dt \right)^{1/p}$$

for a.e. $x, y \in [0,1]$.

5. U denotes an open subset of $\mathbb{R}^n$ with smooth boundary ∂U. All given functions are assumed smooth, unless otherwise stated.

Let U, V be open sets, with $V \Subset U$. Show there exists a smooth function ζ such that $\zeta \equiv 1$ on V, $\zeta = 0$ near ∂U. (Hint: Take $V \Subset W \Subset U$ and mollify χ_W.)

6. U denotes an open subset of $\mathbb{R}^n$ with smooth boundary ∂U. All given functions are assumed smooth, unless otherwise stated.

Assume U is bounded and $U \Subset \bigcup_{i=1}^N V^i$. Show there exist C^∞ functions ζ_i $(i = 1, \dots, N)$ such that

$$\begin{cases} 0 \le \zeta_i \le 1, & \operatorname{spt} \zeta_i \subset V^i \quad (i = 1, \dots, N), \\ \sum_{i=1}^N \zeta_i = 1 & \text{on } U. \end{cases}$$

The functions $\{\zeta_i\}_{i=1}^N$ form a partition of unity.

7. U denotes an open subset of $\mathbb{R}^n$ with smooth boundary ∂U. All given functions are assumed smooth, unless otherwise stated.

Assume that U is bounded and there exists a smooth vector field $\boldsymbol{\alpha}$ such that $\boldsymbol{\alpha} \cdot \boldsymbol{\nu} \geq 1$ along ∂U, where $\boldsymbol{\nu}$ as usual denotes the outward unit normal. Assume $1 \leq p < \infty$.

Apply the Gauss–Green Theorem to $\int_{\partial U} |u|^p \boldsymbol{\alpha} \cdot \boldsymbol{\nu} \, dS$, to derive a new proof of the trace inequality

$$\int_{\partial U} |u|^p \, dS \leq C \int_U |Du|^p + |u|^p \, dx$$

for all $u \in C^1(\bar{U})$.

8. U denotes an open subset of $\mathbb{R}^n$ with smooth boundary ∂U. All given functions are assumed smooth, unless otherwise stated.

Let U be bounded, with a C^1 boundary. Show that a "typical" function $u \in L^p(U)$ $(1 \leq p < \infty)$ does not have a trace on ∂U. More precisely, prove there does not exist a bounded linear operator

$$T : L^p(U) \to L^p(\partial U)$$

such that $Tu = u|_{\partial U}$ whenever $u \in C(\bar{U}) \cap L^p(U)$.

9. U denotes an open subset of $\mathbb{R}^n$ with smooth boundary ∂U. All given functions are assumed smooth, unless otherwise stated.

Integrate by parts to prove the interpolation inequality:

$$\|Du\|_{L^2} \leq C\|u\|_{L^2}^{1/2}\|D^2u\|_{L^2}^{1/2}$$

for all $u \in C_c^\infty(U)$. Assume U is bounded, ∂U is smooth, and prove this inequality if $u \in H^2(U) \cap H_0^1(U)$.

(Hint: Take sequences $\{v_k\}_{k=1}^\infty \subset C_c^\infty(U)$ converging to u in $H_0^1(U)$ and $\{w_k\}_{k=1}^\infty \subset C^\infty(\bar{U})$ converging to u in $H^2(U)$.)

10. U denotes an open subset of $\mathbb{R}^n$ with smooth boundary ∂U. All given functions are assumed smooth, unless otherwise stated.

(a) Integrate by parts to prove

$$\|Du\|_{L^p} \leq C\|u\|_{L^p}^{1/2}\|D^2u\|_{L^p}^{1/2}$$

for $2 \leq p < \infty$ and all $u \in C_c^\infty(U)$.

(Hint: $\int_U |Du|^p\,dx = \sum_{i=1}^n \int_U u_{x_i}u_{x_i}|Du|^{p-2}\,dx$.)

(b) Prove

$$\|Du\|_{L^{2p}} \leq C\|u\|_{L^\infty}^{1/2}\|D^2u\|_{L^p}^{1/2}$$

for $1 \leq p < \infty$ and all $u \in C_c^\infty(U)$.

11. U denotes an open subset of $\mathbb{R}^n$ with smooth boundary ∂U. All given functions are assumed smooth, unless otherwise stated.

Suppose U is connected and $u \in W^{1,p}(U)$ satisfies $Du = 0$ a.e. in U. Prove u is constant a.e. in U.

12. U denotes an open subset of $\mathbb{R}^n$ with smooth boundary ∂U. All given functions are assumed smooth, unless otherwise stated.

Show by example that if we have $\|D^h u\|_{L^1(V)} \leq C$ for all $0 < |h| < \frac{1}{2}\operatorname{dist}(V, \partial U)$, it does not necessarily follow that $u \in W^{1,1}(V)$.

13. U denotes an open subset of $\mathbb{R}^n$ with smooth boundary ∂U. All given functions are assumed smooth, unless otherwise stated.

Give an example of an open set $U \subset \mathbb{R}^n$ and a function $u \in W^{1,\infty}(U)$, such that u is not Lipschitz continuous on U. (Hint: Take U to be the open unit disk in $\mathbb{R}^2$, with a slit removed.)

14. U denotes an open subset of $\mathbb{R}^n$ with smooth boundary ∂U. All given functions are assumed smooth, unless otherwise stated.

Verify that if $n > 1$, the unbounded function $u = \log\log(1 + \frac{1}{|x|})$ belongs to $W^{1,n}(U)$, for $U = B^0(0, 1)$.

15. U denotes an open subset of $\mathbb{R}^n$ with smooth boundary ∂U. All given functions are assumed smooth, unless otherwise stated.

Fix $\alpha > 0$ and let $U = B^0(0,1)$. Show there exists a constant C, depending only on n and α, such that

$$\int_U u^2 \, dx \leq C \int_U |Du|^2 \, dx,$$

provided $|\{x \in U \mid u(x) = 0\}| \geq \alpha$, $u \in H^1(U)$.

16. U denotes an open subset of $\mathbb{R}^n$ with smooth boundary ∂U. All given functions are assumed smooth, unless otherwise stated.

(Variant of Hardy's inequality) Show that for each $n \geq 3$ there exists a constant C so that

$$\int_{\mathbb{R}^n} \frac{u^2}{|x|^2}\, dx \leq C \int_{\mathbb{R}^n} |Du|^2\, dx$$

for all $u \in H^1(\mathbb{R}^n)$.

(Hint: $|Du + \lambda \frac{x}{|x|^2} u|^2 \geq 0$ for each $\lambda \in \mathbb{R}$.)

17. U denotes an open subset of $\mathbb{R}^n$ with smooth boundary ∂U. All given functions are assumed smooth, unless otherwise stated.

(Chain rule) Assume $F : \mathbb{R} \to \mathbb{R}$ is C^1, with F' bounded. Suppose U is bounded and $u \in W^{1,p}(U)$ for some $1 \leq p \leq \infty$. Show

$$v := F(u) \in W^{1,p}(U), \qquad v_{x_i} = F'(u)u_{x_i} \quad (i = 1, \ldots, n).$$

18. U denotes an open subset of $\mathbb{R}^n$ with smooth boundary ∂U. All given functions are assumed smooth, unless otherwise stated.

Assume $1 \leq p \leq \infty$ and U is bounded.

(a) Prove that if $u \in W^{1,p}(U)$, then $|u| \in W^{1,p}(U)$.

(b) Prove $u \in W^{1,p}(U)$ implies $u^+, u^- \in W^{1,p}(U)$, and

$$Du^+ = \begin{cases} Du & \text{a.e. on } \{u > 0\}, \\ 0 & \text{a.e. on } \{u \leq 0\}, \end{cases} \qquad Du^- = \begin{cases} 0 & \text{a.e. on } \{u \geq 0\}, \\ -Du & \text{a.e. on } \{u < 0\}. \end{cases}$$

(Hint: $u^+ = \lim_{\varepsilon \to 0} F_\varepsilon(u)$, for

$$F_\varepsilon(z) := \begin{cases} (z^2 + \varepsilon^2)^{1/2} - \varepsilon & \text{if } z \geq 0, \\ 0 & \text{if } z < 0. \end{cases}$$

)

(c) Prove that if $u \in W^{1,p}(U)$, then $Du = 0$ a.e. on the set $\{u = 0\}$.

19. U denotes an open subset of $\mathbb{R}^n$ with smooth boundary ∂U. All given functions are assumed smooth, unless otherwise stated.

Provide details for the following alternative proof that if $u \in H^1(U)$, then $Du = 0$ a.e. on the set $\{u = 0\}$.

Let ϕ be a smooth, bounded and nondecreasing function, such that ϕ' is bounded and $\phi(z) = z$ if $|z| \leq 1$. Set $u^\epsilon(x) := \epsilon\phi(u/\epsilon)$.

Show that $u^\epsilon \rightharpoonup 0$ weakly in $H^1(U)$ and therefore

$$\int_U Du^\epsilon \cdot Du \, dx = \int_U \phi'(u/\epsilon)|Du|^2 \, dx \to 0.$$

Employ this observation to finish the proof.

20. U denotes an open subset of $\mathbb{R}^n$ with smooth boundary ∂U. All given functions are assumed smooth, unless otherwise stated.

Use the Fourier transform to prove that if $u \in H^s(\mathbb{R}^n)$ for $s > n/2$, then $u \in L^\infty(\mathbb{R}^n)$, with the bound

$$\|u\|_{L^\infty(\mathbb{R}^n)} \leq C\|u\|_{H^s(\mathbb{R}^n)}$$

for a constant C depending only on s and n.

21. U denotes an open subset of $\mathbb{R}^n$ with smooth boundary ∂U. All given functions are assumed smooth, unless otherwise stated.

Show that if $u, v \in H^s(\mathbb{R}^n)$ for $s > n/2$, then $uv \in H^s(\mathbb{R}^n)$ and

$$\|uv\|_{H^s(\mathbb{R}^n)} \leq C\|u\|_{H^s(\mathbb{R}^n)}\|v\|_{H^s(\mathbb{R}^n)},$$

the constant C depending only on s and n.

6. Second-Order Elliptic Equations

1. The coefficients of the PDE are smooth and satisfy uniform ellipticity. $U \subset \mathbb{R}^n$ is open and bounded, with smooth boundary ∂U.

Consider Laplace's equation with potential function c:

$$-\Delta u + cu = 0, \tag{*}$$

and the divergence structure equation:

$$-\operatorname{div}(aDv) = 0, \tag{**}$$

where the function a is positive.

(a) Show that if u solves (*) and $w > 0$ also solves (*), then $v := u/w$ solves (**) for $a := w^2$.

(b) Conversely, show that if v solves (**), then $u := va^{1/2}$ solves (*) for some potential c.

2. The coefficients of the PDE are smooth and satisfy uniform ellipticity. $U \subset \mathbb{R}^n$ is open and bounded, with smooth boundary ∂U.

Let $Lu = -\sum_{i,j=1}^{n}(a^{ij}u_{x_i})_{x_j} + cu$. Prove that there exists a constant $\mu > 0$ such that the corresponding bilinear form $B[\cdot,\cdot]$ satisfies the hypotheses of the Lax–Milgram Theorem, provided

$$c(x) \geq -\mu \qquad (x \in U).$$

3. The coefficients of the PDE are smooth and satisfy uniform ellipticity. $U \subset \mathbb{R}^n$ is open and bounded, with smooth boundary ∂U.

A function $u \in H_0^2(U)$ is a weak solution of this boundary-value problem for the biharmonic equation

$$\begin{cases} \Delta^2 u = f & \text{in } U, \\ u = \frac{\partial u}{\partial \nu} = 0 & \text{on } \partial U, \end{cases} \tag{*}$$

provided $\int_U \Delta u \Delta v \, dx = \int_U f v \, dx$ for all $v \in H_0^2(U)$. Given $f \in L^2(U)$, prove that there exists a unique weak solution of (*).

4. The coefficients of the PDE are smooth and satisfy uniform ellipticity. $U \subset \mathbb{R}^n$ is open and bounded, with smooth boundary ∂U.

Assume U is connected. A function $u \in H^1(U)$ is a weak solution of Neumann's problem

$$\begin{cases} -\Delta u = f & \text{in } U, \\ \frac{\partial u}{\partial \nu} = 0 & \text{on } \partial U, \end{cases} \tag{*}$$

if $\int_U Du \cdot Dv\, dx = \int_U fv\, dx$ for all $v \in H^1(U)$. Suppose $f \in L^2(U)$. Prove (*) has a weak solution if and only if $\int_U f\, dx = 0$.

5. The coefficients of the PDE are smooth and satisfy uniform ellipticity. $U \subset \mathbb{R}^n$ is open and bounded, with smooth boundary ∂U.

Explain how to define $u \in H^1(U)$ to be a weak solution of Poisson's equation with Robin boundary conditions:

$$\begin{cases} -\Delta u = f & \text{in } U, \\ u + \frac{\partial u}{\partial \nu} = 0 & \text{on } \partial U. \end{cases}$$

Discuss the existence and uniqueness of a weak solution for a given $f \in L^2(U)$.

6. The coefficients of the PDE are smooth and satisfy uniform ellipticity. $U \subset \mathbb{R}^n$ is open and bounded, with smooth boundary ∂U.

Suppose U is connected and ∂U consists of two disjoint, closed sets Γ_1 and Γ_2. Define what it means for u to be a weak solution of Poisson's equation with mixed Dirichlet–Neumann boundary conditions:

$$\begin{cases} -\Delta u = f & \text{in } U, \\ u = 0 & \text{on } \Gamma_1, \\ \frac{\partial u}{\partial \nu} = 0 & \text{on } \Gamma_2. \end{cases}$$

Discuss the existence and uniqueness of weak solutions.

7. The coefficients of the PDE are smooth and satisfy uniform ellipticity. $U \subset \mathbb{R}^n$ is open and bounded, with smooth boundary ∂U.

Let $u \in H^1(\mathbb{R}^n)$ have compact support and be a weak solution of the semilinear PDE

$$-\Delta u + c(u) = f \quad \text{in } \mathbb{R}^n,$$

where $f \in L^2(\mathbb{R}^n)$ and $c : \mathbb{R} \to \mathbb{R}$ is smooth, with $c(0) = 0$ and $c' \geq 0$. Assume also $c(u) \in L^2(\mathbb{R}^n)$. Derive the estimate $\|D^2 u\|_{L^2} \leq C\|f\|_{L^2}$.

(Hint: Mimic the proof of Theorem 1 in §6.3.1, but without the cutoff function ζ.)

8. The coefficients of the PDE are smooth and satisfy uniform ellipticity. $U \subset \mathbb{R}^n$ is open and bounded, with smooth boundary ∂U.

Let u be a smooth solution of the uniformly elliptic equation $Lu = -\sum_{i,j=1}^{n} a^{ij}(x) u_{x_i x_j} = 0$ in U. Assume that the coefficients have bounded derivatives.

Set $v := |Du|^2 + \lambda u^2$ and show that $Lv \leq 0$ in U if λ is large enough. Deduce

$$\|Du\|_{L^\infty(U)} \leq C(\|Du\|_{L^\infty(\partial U)} + \|u\|_{L^\infty(\partial U)}).$$

9. The coefficients of the PDE are smooth and satisfy uniform ellipticity. $U \subset \mathbb{R}^n$ is open and bounded, with smooth boundary ∂U.

Assume u is a smooth solution of $Lu = -\sum_{i,j=1}^n a^{ij} u_{x_i x_j} = f$ in U, $u = 0$ on ∂U, where f is bounded. Fix $x^0 \in \partial U$. A barrier at x^0 is a C^2 function w such that $Lw \geq 1$ in U, $w(x^0) = 0$, $w \geq 0$ on ∂U.

Show that if w is a barrier at x^0, there exists a constant C such that

$$|Du(x^0)| \leq C \left| \frac{\partial w}{\partial \nu}(x^0) \right|.$$

10. The coefficients of the PDE are smooth and satisfy uniform ellipticity. $U \subset \mathbb{R}^n$ is open and bounded, with smooth boundary ∂U.

Assume U is connected. Use (a) energy methods and (b) the maximum principle to show that the only smooth solutions of the Neumann boundary-value problem

$$\begin{cases} -\Delta u = 0 & \text{in } U, \\ \frac{\partial u}{\partial \nu} = 0 & \text{on } \partial U \end{cases}$$

are $u \equiv C$, for some constant C.

11. The coefficients of the PDE are smooth and satisfy uniform ellipticity. $U \subset \mathbb{R}^n$ is open and bounded, with smooth boundary ∂U.

Assume $u \in H^1(U)$ is a bounded weak solution of

$$-\sum_{i,j=1}^{n} (a^{ij} u_{x_i})_{x_j} = 0 \quad \text{in } U.$$

Let $\phi : \mathbb{R} \to \mathbb{R}$ be convex and smooth, and set $w = \phi(u)$. Show w is a weak subsolution; that is, $B[w, v] \leq 0$ for all $v \in H_0^1(U)$, $v \geq 0$.

12. The coefficients of the PDE are smooth and satisfy uniform ellipticity. $U \subset \mathbb{R}^n$ is open and bounded, with smooth boundary ∂U.

We say that the uniformly elliptic operator

$$Lu = -\sum_{i,j=1}^{n} a^{ij} u_{x_i x_j} + \sum_{i=1}^{n} b^i u_{x_i} + cu$$

satisfies the weak maximum principle if for all $u \in C^2(U) \cap C(\bar{U})$,

$$\begin{cases} Lu \leq 0 & \text{in } U, \\ u \leq 0 & \text{on } \partial U \end{cases}$$

implies that $u \leq 0$ in U.

Suppose that there exists a function $v \in C^2(U) \cap C(\bar{U})$ such that $Lv \geq 0$ in U and $v > 0$ on $\bar{U}$. Show that L satisfies the weak maximum principle.

(Hint: Find an elliptic operator M with no zeroth-order term such that $w := u/v$ satisfies $Mw \leq 0$ in the region $\{u > 0\}$. To do this, first compute $(v^2 w_{x_i})_{x_j}$.)

13. The coefficients of the PDE are smooth and satisfy uniform ellipticity. $U \subset \mathbb{R}^n$ is open and bounded, with smooth boundary ∂U.

(Courant minimax principle) Let $L = -\sum_{i,j=1}^n (a^{ij} u_{x_i})_{x_j}$, where $((a^{ij}))$ is symmetric. Assume the operator L, with zero boundary conditions, has eigenvalues $0 < \lambda_1 < \lambda_2 \leq \cdots$. Show

$$\lambda_k = \max_{S \in \Sigma_{k-1}} \min_{\substack{u \in S^\perp \\ \|u\|_{L^2}=1}} B[u,u] \qquad (k = 1, 2, \ldots).$$

Here Σ_{k-1} denotes the collection of $(k-1)$-dimensional subspaces of $H_0^1(U)$.

14. The coefficients of the PDE are smooth and satisfy uniform ellipticity. $U \subset \mathbb{R}^n$ is open and bounded, with smooth boundary ∂U.

Let λ_1 be the principal eigenvalue of the uniformly elliptic, nonsymmetric operator

$$Lu = -\sum_{i,j=1}^{n} a^{ij} u_{x_i x_j} + \sum_{i=1}^{n} b^i u_{x_i} + cu,$$

taken with zero boundary conditions. Prove the "max-min" representation formula:

$$\lambda_1 = \sup_u \inf_x \frac{Lu(x)}{u(x)},$$

the "sup" taken over functions $u \in C^\infty(\bar{U})$ with $u > 0$ in U, $u = 0$ on ∂U, and the "inf" taken over points $x \in U$.

(Hint: Consider the eigenfunction w_1^* corresponding to λ_1 for the adjoint operator L^*.)

15. The coefficients of the PDE are smooth and satisfy uniform ellipticity. $U \subset \mathbb{R}^n$ is open and bounded, with smooth boundary ∂U.

(Eigenvalues and domain variations) Consider a family of smooth, bounded domains $U(\tau) \subset \mathbb{R}^n$ that depend smoothly upon the parameter $\tau \in \mathbb{R}$. As τ changes, each point on $\partial U(\tau)$ moves with velocity $\mathbf{v}$.

For each τ, we consider eigenvalues $\lambda = \lambda(\tau)$ and corresponding eigenfunctions $w = w(x, \tau)$:

$$\begin{cases} -\Delta w = \lambda w & \text{in } U(\tau), \\ w = 0 & \text{on } \partial U(\tau), \end{cases}$$

normalized so that $\|w\|_{L^2(U(\tau))} = 1$. Suppose that λ and w are smooth functions of τ and x.

Prove Hadamard's variational formula

$$\dot{\lambda} = -\int_{\partial U(\tau)} \left| \frac{\partial w}{\partial \nu} \right|^2 \mathbf{v} \cdot \nu \, dS,$$

where $\dot{} = \frac{d}{d\tau}$ and $\mathbf{v} \cdot \nu$ is the normal velocity of $\partial U(\tau)$.

(Hint: Use the calculus formula from §C.4.)

16. The coefficients of the PDE are smooth and satisfy uniform ellipticity. $U \subset \mathbb{R}^n$ is open and bounded, with smooth boundary ∂U.

(Radiation condition) If we separate variables to look for a complex-valued solution of the wave equation having the form $u = e^{-i\sigma t}w$ for $w = w(x)$ and $\sigma \in \mathbb{R}$, $\sigma \neq 0$, we are led to the eigenvalue problem

$$-\Delta w = \lambda w \quad \text{in } \mathbb{R}^n \tag{*}$$

where $\lambda := \sigma^2$.

(a) Show that $w = e^{i\sigma\omega\cdot x}$ solves (*), provided $|\omega| = 1$. Then $u = e^{i\sigma(\omega\cdot x - t)}$ is a traveling wave solution of the wave equation.

(b) Show that for $n = 3$, the function $\Phi := \frac{e^{i\sigma|x|}}{4\pi|x|}$ solves $-\Delta\Phi = \lambda\Phi + \delta_0$ in $\mathbb{R}^3$.

(c) The Sommerfeld radiation condition requires for a solution of (*) that

$$\lim_{r\to\infty} r(w_r - i\sigma w) = 0,$$

for $w_r := Dw \cdot \frac{x}{|x|}$. Prove that the solution w from (a) does not satisfy this condition but that Φ from (b) does.

17. The coefficients of the PDE are smooth and satisfy uniform ellipticity. $U \subset \mathbb{R}^n$ is open and bounded, with smooth boundary ∂U.

(Continuation) Prove that if w is a complex-valued solution of eigenvalue problem (*) in $\mathbb{R}^3$ and if w satisfies the radiation condition, then $w \equiv 0$.

(Hints: First observe that

$$0 = \int_{B(0,R)} \bar{w}\Delta w - w\Delta\bar{w}\, dx = \int_{\partial B(0,R)} \bar{w}w_r - w\bar{w}_r\, dS.$$

Use this and the radiation condition to show

$$\int_{\partial B(0,R)} |w_r|^2 + \sigma^2|w|^2\, dS = \int_{\partial B(0,R)} |w_r - i\sigma w|^2\, dS \to 0$$

as $R \to \infty$. Given now a point $x^0 \in \mathbb{R}^3$, select $R > |x^0|$. Then

$$w(x^0) = \int_{\partial B(0,R)} \Phi w_r - w\Phi_r\, dS,$$

where $\Phi = \Phi(x - x^0)$. Show the integral goes to zero as $R \to \infty$.)

7. Linear Evolution Equations

1. $U \subset \mathbb{R}^n$ is open and bounded, with smooth boundary, and $T > 0$.

Prove there is at most one smooth solution of this initial/boundary-value problem for the heat equation with Neumann boundary conditions

$$\begin{cases} u_t - \Delta u = f & \text{in } U_T, \\ \frac{\partial u}{\partial \nu} = 0 & \text{on } \partial U \times [0,T], \\ u = g & \text{on } U \times \{t = 0\}. \end{cases}$$

2. $U \subset \mathbb{R}^n$ is open and bounded, with smooth boundary, and $T > 0$.

Assume u is a smooth solution of

$$\begin{cases} u_t - \Delta u = 0 & \text{in } U \times (0, \infty), \\ u = 0 & \text{on } \partial U \times [0, \infty), \\ u = g & \text{on } U \times \{t = 0\}. \end{cases}$$

Prove the exponential decay estimate:

$$\|u(\cdot, t)\|_{L^2(U)} \leq e^{-\lambda_1 t} \|g\|_{L^2(U)} \qquad (t \geq 0),$$

where $\lambda_1 > 0$ is the principal eigenvalue of $-\Delta$ (with zero boundary conditions) on U.

3. $U \subset \mathbb{R}^n$ is open and bounded, with smooth boundary, and $T > 0$.

(Adjoint dynamics) Suppose that u is a smooth solution of

$$\begin{cases} u_t + Lu = 0 & \text{in } U_T, \\ u = 0 & \text{on } \partial U \times [0,T], \\ u = g & \text{on } U \times \{t = 0\}, \end{cases}$$

where L denotes a second-order elliptic operator, and that v is a smooth solution of the adjoint problem

$$\begin{cases} v_t - L^* v = 0 & \text{in } U_T, \\ v = 0 & \text{on } \partial U \times [0,T], \\ v = h & \text{on } U \times \{t = T\}. \end{cases}$$

Show $\int_U v(\cdot,0) g \, dx = \int_U u(\cdot,T) h \, dx$.

4. $U \subset \mathbb{R}^n$ is open and bounded, with smooth boundary, and $T > 0$.

(Galerkin's method for Poisson's equation) Suppose $f \in L^2(U)$ and assume that $u_m = \sum_{k=1}^m d_m^k w_k$ solves

$$\int_U Du_m \cdot Dw_k \, dx = \int_U f \cdot w_k \, dx$$

for $k = 1, \dots, m$, where $\{w_k\}_{k=1}^\infty$ is an orthogonal basis of $H_0^1(U)$. Show that $\{u_m\}_{m=1}^\infty$ converges weakly in $H_0^1(U)$ to the weak solution u of

$$\begin{cases} -\Delta u = f & \text{in } U, \\ u = 0 & \text{on } \partial U. \end{cases}$$

5. $U \subset \mathbb{R}^n$ is open and bounded, with smooth boundary, and $T > 0$.

Assume

$$\begin{cases} \mathbf{u}_k \rightharpoonup \mathbf{u} & \text{in } L^2(0,T;H_0^1(U)), \\ \mathbf{u}_k' \rightharpoonup \mathbf{v} & \text{in } L^2(0,T;H^{-1}(U)). \end{cases}$$

Prove that $\mathbf{v} = \mathbf{u}'$. (Hint: Let $\phi \in C_c^1(0,T)$, $w \in H_0^1(U)$. Then

$$\int_0^T \langle \mathbf{u}_k', \phi w \rangle \, dt = -\int_0^T \langle \mathbf{u}_k, \phi' w \rangle \, dt.$$

)

6. $U \subset \mathbb{R}^n$ is open and bounded, with smooth boundary, and $T > 0$.

Suppose H is a Hilbert space and $\mathbf{u}_k \rightharpoonup \mathbf{u}$ in $L^2(0,T;H)$. Assume further we have the uniform bounds

$$\operatorname*{ess\,sup}_{0\le t\le T} \|\mathbf{u}_k(t)\| \le C \qquad (k = 1, \ldots)$$

for some constant C. Prove $\operatorname{ess\,sup}_{0\le t\le T} \|\mathbf{u}(t)\| \le C$.

(Hint: We have $\int_a^b (v, \mathbf{u}_k(t))\, dt \le C\|v\||b-a|$ for $0 \le a \le b \le T$ and $v \in H$.)

7. $U \subset \mathbb{R}^n$ is open and bounded, with smooth boundary, and $T > 0$.

Suppose u is a smooth solution of

$$\begin{cases} u_t - \Delta u + cu = 0 & \text{in } U \times (0, \infty), \\ u = 0 & \text{on } \partial U \times [0, \infty), \\ u = g & \text{on } U \times \{t = 0\}, \end{cases}$$

and the function c satisfies $c \geq \gamma > 0$. Prove the pointwise exponential decay estimate

$$|u(x,t)| \leq Ce^{-\gamma t} \qquad ((x,t) \in U \times (0, \infty)).$$

8. $U \subset \mathbb{R}^n$ is open and bounded, with smooth boundary, and $T > 0$.

Assume that u is a smooth solution of the PDE from Problem 7, that $g \geq 0$, and that c is bounded (but not necessarily nonnegative). Show $u \geq 0$.

(Hint: What PDE does $v := e^{-\lambda t}u$ solve?)

9. $U \subset \mathbb{R}^n$ is open and bounded, with smooth boundary, and $T > 0$.

Prove inequality (54) in §7.1.3, (59) in §7.2.3.

(Hints: Assume u is smooth, $u = 0$ on ∂U. Transform the term $(Lu, -\Delta u)$ by integrating by parts twice, and then estimate the boundary terms. After changing variables locally and using cutoff functions, you may assume the boundary is flat. This problem is difficult.)

10. $U \subset \mathbb{R}^n$ is open and bounded, with smooth boundary, and $T > 0$.

Show there exists at most one smooth solution of this initial/boundary-value problem for the telegraph equation

$$\begin{cases} u_{tt} + du_t - u_{xx} = f & \text{in } (0,1) \times (0,T), \\ u = 0 & \text{on } (\{0\} \times [0,T]) \cup (\{1\} \times [0,T]), \\ u = g,\ u_t = h & \text{on } (0,1) \times \{t = 0\}. \end{cases}$$

Here d is a constant.

11. $U \subset \mathbb{R}^n$ is open and bounded, with smooth boundary, and $T > 0$.

Prove there exists at most one smooth solution u of this problem for the beam equation

$$\begin{cases} u_{tt} + u_{xxxx} = 0 & \text{in } (0,1) \times (0,T), \\ u = u_x = 0 & \text{on } (\{0\} \times [0,T]) \cup (\{1\} \times [0,T]), \\ u = g,\ u_t = h & \text{on } [0,1] \times \{t = 0\}. \end{cases}$$

12. $U \subset \mathbb{R}^n$ is open and bounded, with smooth boundary, and $T > 0$.

Prove the resolvent identities (12) and (13) in §7.4.1.

13. $U \subset \mathbb{R}^n$ is open and bounded, with smooth boundary, and $T > 0$.

Justify the equality

$$A \int_0^\infty e^{-\lambda t} S(t)u \, dt = \int_0^\infty e^{-\lambda t} AS(t)u \, dt$$

used in (16) of §7.4.1 when $u \in D(A)$. (Hint: Approximate the integral by a Riemann sum and recall A is a closed operator.)

14. $U \subset \mathbb{R}^n$ is open and bounded, with smooth boundary, and $T > 0$.

Define for $t > 0$

$$[S(t)g](x) = \int_{\mathbb{R}^n} \Phi(x-y,t)g(y)\,dy \qquad (x \in \mathbb{R}^n),$$

where $g : \mathbb{R}^n \to \mathbb{R}$ and Φ is the fundamental solution of the heat equation. Also set $S(0)g = g$.

(a) Prove $\{S(t)\}_{t\geq 0}$ is a contraction semigroup on $L^2(\mathbb{R}^n)$.

(b) Show $\{S(t)\}_{t\geq 0}$ is not a contraction semigroup on $L^\infty(\mathbb{R}^n)$.

15. $U \subset \mathbb{R}^n$ is open and bounded, with smooth boundary, and $T > 0$.

Let $\{S(t)\}_{t\geq 0}$ be a contraction semigroup on X, with generator A. Inductively define $D(A^k) := \{u \in D(A^{k-1}) \mid A^{k-1}u \in D(A)\}$ $(k = 2, \ldots)$. Show that if $u \in D(A^k)$ for some k, then $S(t)u \in D(A^k)$ for each $t \geq 0$.

16. $U \subset \mathbb{R}^n$ is open and bounded, with smooth boundary, and $T > 0$.

Use Problem 15 to prove that if u is the semigroup solution in $X = L^2(U)$ of

$$\begin{cases} u_t - \Delta u = 0 & \text{in } U_T, \\ u = 0 & \text{on } \partial U \times [0,T], \\ u = g & \text{on } U \times \{t = 0\}, \end{cases}$$

with $g \in C_c^\infty(U)$, then $u(\cdot, t) \in C^\infty(U)$ for each $0 \le t \le T$.

8. The Calculus of Variations

1. U denotes a bounded, open subset of $\mathbb{R}^n$ with smooth boundary. All given functions are assumed smooth, unless otherwise stated.

This problem illustrates that a weakly convergent sequence can be badly behaved.

(a) Prove $u_k(x) = \sin(kx) \rightharpoonup 0$ as $k \to \infty$ in $L^2(0,1)$.

(b) Fix $a, b \in \mathbb{R}$, $0 < \lambda < 1$. Define

$$u_k(x) := \begin{cases} a & \text{if } j/k \le x < (j+\lambda)/k, \\ b & \text{if } (j+\lambda)/k \le x < (j+1)/k \end{cases} \qquad (j = 0, \dots, k-1).$$

Prove $u_k \rightharpoonup \lambda a + (1-\lambda)b$ in $L^2(0,1)$.

2. U denotes a bounded, open subset of $\mathbb{R}^n$ with smooth boundary. All given functions are assumed smooth, unless otherwise stated.

Let $\phi : U \to \mathbb{R}$. Find $L = L(p, z, x)$ so that the PDE $-\Delta u + D\phi \cdot Du = f$ in U is the Euler–Lagrange equation corresponding to the functional $I[w] := \int_U L(Dw, w, x)\, dx$.

(Hint: Look for a Lagrangian with an exponential term.)

3. U denotes a bounded, open subset of $\mathbb{R}^n$ with smooth boundary. All given functions are assumed smooth, unless otherwise stated.

The elliptic regularization of the heat equation is the PDE

$$u_t - \Delta u - \epsilon u_{tt} = 0 \quad \text{in } U_T, \tag{*}$$

where $\epsilon > 0$ and $U_T = U \times (0, T]$. Show that (*) is the Euler–Lagrange equation corresponding to an energy functional

$$I_\varepsilon[w] := \iint_{U_T} L_\varepsilon(Dw, w_t, w, x, t)\, dx\, dt.$$

(Hint: Look for a Lagrangian with an exponential term involving t.)

4. U denotes a bounded, open subset of $\mathbb{R}^n$ with smooth boundary. All given functions are assumed smooth, unless otherwise stated.

Assume $\eta : \mathbb{R}^n \to \mathbb{R}$ is C^1.

(a) Show $L(P, z, x) = \eta(z) \det P$ ($P \in \mathbb{M}^{n\times n}$, $z \in \mathbb{R}^n$) is a null Lagrangian.

(b) Deduce that if $\mathbf{u} : \mathbb{R}^n \to \mathbb{R}^n$ is C^2, then $\int_U \eta(\mathbf{u}) \det D\mathbf{u}\, dx$ depends only on $\mathbf{u}|_{\partial U}$.

5. U denotes a bounded, open subset of $\mathbb{R}^n$ with smooth boundary. All given functions are assumed smooth, unless otherwise stated.

(Continuation) Fix $x^0 \notin \mathbf{u}(\partial U)$, and choose a function η as above so that $\int_{\mathbb{R}^n} \eta \, dz = 1$, $\operatorname{spt} \eta \subset B(x^0, r)$, r taken so small that $B(x^0, r) \cap \mathbf{u}(\partial U) = \varnothing$. Define

$$\deg(\mathbf{u}, x^0) = \int_U \eta(\mathbf{u}) \det D\mathbf{u} \, dx,$$

the degree of $\mathbf{u}$ relative to x^0. Prove the degree is an integer.

6. U denotes a bounded, open subset of $\mathbb{R}^n$ with smooth boundary. All given functions are assumed smooth, unless otherwise stated.

Let $\Sigma \subset \mathbb{R}^3$ denote the graph of the smooth function $u : U \to \mathbb{R}$, $U \subset \mathbb{R}^2$. Then

$$\int_U (1 + |Du|^2)^{-3/2} \det D^2 u \, dx \qquad (*)$$

represents the integral of the Gauss curvature over Σ. Prove that this expression depends only upon Du restricted to ∂U. (The Gauss–Bonnet Theorem in differential geometry computes (*) in terms of the geodesic curvature of $\partial\Sigma$.)

7. U denotes a bounded, open subset of $\mathbb{R}^n$ with smooth boundary. All given functions are assumed smooth, unless otherwise stated.

Let $m = n$. Prove $L(P) = \mathrm{tr}(P^2) - \mathrm{tr}(P)^2$ $(P \in \mathbb{M}^{n\times n})$ is a null Lagrangian.

8. U denotes a bounded, open subset of $\mathbb{R}^n$ with smooth boundary. All given functions are assumed smooth, unless otherwise stated.

Explain why the methods in §8.2 will not work to prove the existence of a minimizer of the functional

$$I[w] := \int_U (1 + |Dw|^2)^{1/2}\, dx$$

over $\mathcal{A} := \{w \in W^{1,q}(U) \mid w = g \text{ on } \partial U\}$, for any $1 \leq q < \infty$.

9. U denotes a bounded, open subset of $\mathbb{R}^n$ with smooth boundary. All given functions are assumed smooth, unless otherwise stated.

(Second variation for systems) Assume $\mathbf{u} : U \to \mathbb{R}^m$ is a smooth minimizer of the functional $I[\mathbf{w}] := \int_U L(D\mathbf{w}, \mathbf{w}, x)\, dx$.

(a) Show

$$\sum_{i,j=1}^{n} \sum_{k,l=1}^{m} \frac{\partial^2 L}{\partial p_i^k \partial p_j^l}(D\mathbf{u}, \mathbf{u}, x)\eta^k \eta^l \xi_i \xi_j \geq 0$$

for all $x \in U$, $\xi \in \mathbb{R}^n$, $\eta \in \mathbb{R}^m$.

(b) Provide an example of a nonconvex function $L : \mathbb{M}^{m\times n} \to \mathbb{R}$ satisfying

$$\sum_{i,j=1}^{n} \sum_{k,l=1}^{m} \frac{\partial^2 L(P)}{\partial p_i^k \partial p_j^l}\eta^k \eta^l \xi_i \xi_j \geq 0$$

for all $P \in \mathbb{M}^{m\times n}$, $\xi \in \mathbb{R}^n$, $\eta \in \mathbb{R}^m$.

10. U denotes a bounded, open subset of $\mathbb{R}^n$ with smooth boundary. All given functions are assumed smooth, unless otherwise stated.

Use the methods of §8.4.1 to show the existence of a nontrivial weak solution $u \in H_0^1(U)$, $u \not\equiv 0$, of

$$\begin{cases} -\Delta u = |u|^{q-1}u & \text{in } U, \\ u = 0 & \text{on } \partial U \end{cases}$$

for $1 < q < \frac{n+2}{n-2}$, $n > 2$.

11. U denotes a bounded, open subset of $\mathbb{R}^n$ with smooth boundary. All given functions are assumed smooth, unless otherwise stated.

Assume $\beta : \mathbb{R} \to \mathbb{R}$ is smooth, with $0 < a \leq \beta'(z) \leq b$ $(z \in \mathbb{R})$ for constants a, b. Let $f \in L^2(U)$. Formulate what it means for $u \in H^1(U)$ to be a weak solution of the nonlinear boundary-value problem

$$\begin{cases} -\Delta u = f & \text{in } U, \\ \frac{\partial u}{\partial \nu} + \beta(u) = 0 & \text{on } \partial U. \end{cases}$$

Prove there exists a unique weak solution.

12. U denotes a bounded, open subset of $\mathbb{R}^n$ with smooth boundary. All given functions are assumed smooth, unless otherwise stated.

Assume u is a smooth minimizer of the area integral

$$I[w] = \int_U (1 + |Dw|^2)^{1/2}\, dx,$$

subject to given boundary conditions $w = g$ on ∂U and the constraint $J[w] = \int_U w\, dx = 1$.

Prove the graph of u is a surface of constant mean curvature.

(Hint: Recall Example 4 in §8.1.2.)

13. U denotes a bounded, open subset of $\mathbb{R}^n$ with smooth boundary. All given functions are assumed smooth, unless otherwise stated.

Assume $f \in L^2(U)$. Prove the dual variational principle that

$$\min_{w \in H_0^1(U)} \int_U \tfrac{1}{2}|Dw|^2 - fw \, dx = \max_{\substack{\boldsymbol{\xi} \in L^2(U;\mathbb{R}^n) \\ \operatorname{div} \boldsymbol{\xi} = f}} -\frac{1}{2} \int_U |\boldsymbol{\xi}|^2 \, dx.$$

14. U denotes a bounded, open subset of $\mathbb{R}^n$ with smooth boundary. All given functions are assumed smooth, unless otherwise stated.

(Multivalued PDE) Show that the variational inequality (26) for the obstacle problem in §8.4.2 can be rewritten as $-\Delta u + \beta(u-h) \ni f$ for the multivalued function

$$\beta(z) := \begin{cases} 0 & \text{if } z > 0, \\ (-\infty, 0] & \text{if } z = 0, \\ \varnothing & \text{if } z < 0. \end{cases}$$

(See also Problem 3 in Chapter 9.)

15. U denotes a bounded, open subset of $\mathbb{R}^n$ with smooth boundary. All given functions are assumed smooth, unless otherwise stated.

(Pointwise gradient constraint)

(a) Show there exists a unique minimizer $u \in \mathcal{A}$ of $I[w] := \int_U \frac{1}{2}|Dw|^2 - fw\,dx$, where $f \in L^2(U)$ and

$$\mathcal{A} := \{w \in H_0^1(U) \mid |Dw| \leq 1 \text{ a.e.}\}.$$

(b) Prove $\int_U Du \cdot D(w-u)\,dx \geq \int_U f(w-u)\,dx$ for all $w \in \mathcal{A}$.

16. U denotes a bounded, open subset of $\mathbb{R}^n$ with smooth boundary. All given functions are assumed smooth, unless otherwise stated.

Assume $n \geq 3$ and U is a bounded open set containing 0. Show that $\mathbf{u} := \frac{x}{|x|}$ belongs to $H^1(U; \mathbb{R}^n)$ and is a harmonic mapping into the sphere S^{n-1}. That is, show $\mathbf{u}$ is a weak solution of

$$\begin{cases} -\Delta \mathbf{u} = |D\mathbf{u}|^2 \mathbf{u}, \\ |\mathbf{u}| = 1 \end{cases} \qquad \text{in } U.$$

17. U denotes a bounded, open subset of $\mathbb{R}^n$ with smooth boundary. All given functions are assumed smooth, unless otherwise stated.

Let $u, \hat{u} \in H_0^1(U)$ both be minimizers of the Dirichlet energy $I[w] := \int_U |Dw|^2\,dx$, subject to $\int_U w^2\,dx = 1$.

Suppose also that $u, \hat{u} > 0$ within U. Follow the hints to give a new proof that $u \equiv \hat{u}$ in U.

(Hint: Define $w = (\frac{u^2+\hat{u}^2}{2})^{1/2}$, $s = \frac{u^2}{u^2+\hat{u}^2}$ and $\eta = \frac{u^2+\hat{u}^2}{2}$; and show that

$$|Dw|^2 = \eta \left| s\frac{Du}{u} + (1-s)\frac{D\hat{u}}{\hat{u}} \right|^2.$$

Deduce

$$\begin{aligned} |Dw|^2 &\le \eta \left(s \left| \frac{Du}{u} \right|^2 + (1-s) \left| \frac{D\hat{u}}{\hat{u}} \right|^2 \right) \\ &= \tfrac{1}{2}|Du|^2 + \tfrac{1}{2}|D\hat{u}|^2 \end{aligned}$$

and therefore $\frac{Du}{u} = \frac{D\hat{u}}{\hat{u}}$ almost everywhere.)

(Belloni–Kawohl, *Manuscripta Math.* 109 (2002), 229–231)

18. U denotes a bounded, open subset of $\mathbb{R}^n$ with smooth boundary. All given functions are assumed smooth, unless otherwise stated.

Assume that a_1, a_2 are smooth, positive functions on $\bar{U}$ such that $a_1 \leq a_2$. Let u_1, u_2 be smooth solutions of

$$\operatorname{div}(a_1 Du_1) = 0, \quad \operatorname{div}(a_2 Du_2) = 0 \quad \text{in } U$$

with $Du_2 \neq 0$ a.e. Suppose finally that $u_1 = u_2$, $\frac{\partial u_1}{\partial \nu} = \frac{\partial u_2}{\partial \nu}$, $a_1 = a_2$ on ∂U.

Prove that $a_1 \equiv a_2$, $u_1 \equiv u_2$ within U.

(Hint: Observe that $\int_U a_1 |Du_1|^2 \, dx = \int_U a_2 |Du_2|^2 \, dx$.)

19. U denotes a bounded, open subset of $\mathbb{R}^n$ with smooth boundary. All given functions are assumed smooth, unless otherwise stated.

(Momentum conservation) Given a solution u of the nonlinear wave equation $\Box u + f(u) = 0$, apply Noether's Theorem to the transformations $\mathbf{x}(x,t,\tau) = (x + \tau e_k, t)$, $w(x,t,\tau) = u(x + \tau e_k, t)$ to calculate the momentum density p_k and the momentum flux $\mathbf{j}_k$ satisfying the conservation laws

$$(p_k)_t - \operatorname{div}(\mathbf{j}_k) = 0 \qquad (k = 1, \dots, n).$$

20. U denotes a bounded, open subset of $\mathbb{R}^n$ with smooth boundary. All given functions are assumed smooth, unless otherwise stated.

Let u be harmonic in some region $U \subseteq \mathbb{R}^n$ and assume $B(0,R) \subset U$, $u(0) = 0$, $u \not\equiv 0$. Define for $0 < r < R$ the functions

$$a(r) := \frac{1}{r^{n-1}} \int_{\partial B(0,r)} u^2 \, dS, \qquad b(r) := \frac{1}{r^{n-2}} \int_{B(0,r)} |Du|^2 \, dx.$$

We derived in §8.6.2 the monotonicity formula $b' = \frac{2}{r^{n-2}} \int_{\partial B(0,r)} u_r^2 \, dS$.

(a) Prove that $a' = \frac{2}{r^{n-1}} \int_{\partial B(0,r)} u u_r \, dS = \frac{2}{r} b$.

(b) Show $b^2 \leq \frac{r}{2} a b'$.

(c) Define the frequency function $f := b/a$ and derive Almgren's monotonicity formula: $f' \geq 0$.

(d) Demonstrate next that $a'/a \leq \beta/r$ and consequently $a(r) \geq \gamma r^\beta$ $(0 < r < R)$ for $\beta := \frac{2b(R)}{a(R)}$ and $\gamma := \frac{a(R)}{R^\beta}$. This is an estimate from below on how fast a nonconstant harmonic function must grow near a point where it vanishes.

9. Nonvariational Techniques

1. U denotes a bounded, open subset of $\mathbb{R}^n$ with smooth boundary.

Assume the vector field $\mathbf{v}$ is smooth. Give another proof of the lemma in §9.1 for this case by solving the ODE

$$\begin{cases} \dot{\mathbf{x}}(t) = -\mathbf{v}(\mathbf{x}(t)) \quad (t \geq 0), \\ \mathbf{x}(0) = y. \end{cases}$$

Let us write the solution as $\mathbf{x}(t, y)$ to display the dependence on the initial point y. For each fixed time $t > 0$, the map $y \mapsto \mathbf{x}(t, y)$ is continuous and so has a fixed point. Conclude that $\mathbf{v}$ has a zero in the closed ball $B(0, r)$.

2. U denotes a bounded, open subset of $\mathbb{R}^n$ with smooth boundary.

Assume $a : \mathbb{R} \to \mathbb{R}$ is continuous and $a(f_n) \rightharpoonup a(f)$ weakly in $L^2(0,1)$ whenever $f_n \rightharpoonup f$ weakly in $L^2(0,1)$. Show a is an affine function; that is, a has the form $a(z) = \alpha z + \beta$ $(z \in \mathbb{R})$ for constants α, β.

3. U denotes a bounded, open subset of $\mathbb{R}^n$ with smooth boundary.

(Penalty method) Let $\epsilon > 0$. Define

$$\beta_\epsilon(z) := \begin{cases} 0 & \text{if } z \geq 0, \\ z/\epsilon & \text{if } z \leq 0, \end{cases}$$

and suppose $u^\epsilon \in H_0^1(U)$ is the weak solution of

$$\begin{cases} -\Delta u^\epsilon + \beta_\epsilon(u^\epsilon) = f & \text{in } U, \\ u^\epsilon = 0 & \text{on } \partial U, \end{cases} \tag{*}$$

where $f \in L^2(U)$. Prove that as $\epsilon \to 0$, $u^\epsilon \rightharpoonup u$ weakly in $H_0^1(U)$, u being the unique nonnegative solution of the variational inequality

$$\int_U Du \cdot D(w-u)\,dx \geq \int_U f(w-u)\,dx$$

for all $w \in H_0^1(U)$ with $w \geq 0$ a.e. Approximating the variational inequality by (*) is the penalty method.

4. U denotes a bounded, open subset of $\mathbb{R}^n$ with smooth boundary.

(Solutions periodic in time) Assume

$$\begin{cases} u_t - \Delta u = f & \text{in } U \times (0, \infty), \\ u = 0 & \text{on } \partial U \times (0, \infty), \\ u = g & \text{on } U \times \{t = 0\}, \end{cases}$$

where $g \in L^2(U)$, $f \in L^\infty(U \times (0, \infty))$. Suppose $\tau > 0$ and f is τ-periodic in t; that is, $f(x, t) = f(x, t + \tau)$ $(x \in U,\ t \geq 0)$. Prove there exists a unique function $g \in L^2(U)$ for which the corresponding solution u is τ-periodic.

5. U denotes a bounded, open subset of $\mathbb{R}^n$ with smooth boundary.

Consider the nonlinear boundary-value problem

$$\begin{cases} -\Delta u + b(Du) = f & \text{in } U, \\ u = 0 & \text{on } \partial U. \end{cases}$$

Use Banach's Fixed Point Theorem to show there exists a unique weak solution $u \in H^2(U) \cap H_0^1(U)$ provided $b : \mathbb{R}^n \to \mathbb{R}$ is Lipschitz continuous, with $\mathrm{Lip}(b)$ small enough.

6. U denotes a bounded, open subset of $\mathbb{R}^n$ with smooth boundary.

Assume $f : \mathbb{R} \to \mathbb{R}$ is Lipschitz continuous, bounded, with $f(0) = 0$ and $f'(0) > \lambda_1$, λ_1 denoting the principal eigenvalue for $-\Delta$ on $H_0^1(U)$. Use the method of sub- and supersolutions to show there exists a weak solution u of

$$\begin{cases} -\Delta u = f(u) & \text{in } U, \\ u = 0 & \text{on } \partial U, \\ u > 0 & \text{in } U. \end{cases}$$

7. U denotes a bounded, open subset of $\mathbb{R}^n$ with smooth boundary.

Assume that $\underline{u}, \bar{u}$ are smooth sub- and supersolutions of the boundary-value problem (1) in §9.3. Use the maximum principle to verify directly

$$\underline{u} = u_0 \leq u_1 \leq \cdots \leq u_k \leq \cdots \leq \bar{u},$$

where the $\{u_k\}_{k=0}^{\infty}$ are defined as in §9.3.

8. U denotes a bounded, open subset of $\mathbb{R}^n$ with smooth boundary.

(Noncompact families of solutions)

(a) Assume $n \geq 3$. Find a constant c such that $u(x) := (1+|x|^2)^{(2-n)/2}$ solves Yamabe's equation

$$-\Delta u = cu^{\frac{n+2}{n-2}} \quad \text{in } \mathbb{R}^n.$$

Note the appearance of the critical exponent $\frac{n+2}{n-2}$. (Compare with Problem 6 in Chapter 4.)

(b) Check that for each $\lambda > 0$, $u_\lambda(x) := (\frac{\lambda}{\lambda^2+|x|^2})^{(n-2)/2}$ is also a solution.

(c) Show that

$$\|u_\lambda\|_{L^{2^*}(\mathbb{R}^n)} = \|u\|_{L^{2^*}(\mathbb{R}^n)}, \quad \|Du_\lambda\|_{L^2(\mathbb{R}^n)} = \|Du\|_{L^2(\mathbb{R}^n)}$$

for each λ and thus that $\{u_\lambda\}_{\lambda>0}$ is not precompact in $L^{2^*}(\mathbb{R}^n)$.

9. U denotes a bounded, open subset of $\mathbb{R}^n$ with smooth boundary.

Let u solve the semilinear heat equation $u_t - \Delta u = f(u)$ in $\mathbb{R}^n \times (0, \infty)$. Assume that u and its derivatives go to zero rapidly as $|x| \to \infty$.

(a) Show that

$$\frac{d}{dt} \int_{\mathbb{R}^n} \tfrac{1}{2}|Du|^2 - F(u)\, dx = -\int_{\mathbb{R}^n} u_t^2\, dx,$$

where $F' = f$, $F(0) = 0$.

(b) Now show that

$$\begin{aligned} &\frac{d}{dt} \int_{\mathbb{R}^n} |x|^2 (\tfrac{1}{2}|Du|^2 - F(u))\, dx \\ &\quad = -\int_{\mathbb{R}^n} u_t^2 |x|^2 - 2nF(u) + (n-2)|Du|^2\, dx. \end{aligned}$$

This is a parabolic analogue of the Derrick–Pohozaev identity. (Bauman, Chen, Phillips, *Euro. J. Applied Math.* 6 (1995), 115–126)

10. U denotes a bounded, open subset of $\mathbb{R}^n$ with smooth boundary.

Let $K \subset \mathbb{R}^n$ be a closed, convex, nonempty set. Define $I[x] := 0$ if $x \in K$ and $I[x] := \infty$ if $x \notin K$. Explicitly determine $A = \partial I$, $J_\lambda = (I + \lambda A)^{-1}$, $A_\lambda = \frac{I - J_\lambda}{\lambda}$ $(\lambda > 0)$ in terms of the geometry of K.

11. U denotes a bounded, open subset of $\mathbb{R}^n$ with smooth boundary.

Give a simple example showing that the flow

$$\mathbf{u}' \in -\partial I[\mathbf{u}] \qquad (t > 0) \tag{*}$$

may be irreversible. (That is, find a Hilbert space H and a convex, proper, lower semicontinuous function $I : H \to (-\infty, +\infty]$ such that the semigroup solution of (*) satisfies $S(t)u = S(t)\hat{u}$ for some $t > 0$ and $u \neq \hat{u}$.)

12. U denotes a bounded, open subset of $\mathbb{R}^n$ with smooth boundary.

Let $u = u(x,t)$ denote the height at $x \in \mathbb{R}^2$ of a sandpile that grows as sand is added at rate $f = f(x,t) \geq 0$. We assume the stability condition $|Du| \leq 1$, meaning that nowhere can the sandpile have slope greater than 1. As usual, $D_x u = Du$. We propose the dynamics

$$u_t - \operatorname{div}(aDu) = f \quad \text{in } \mathbb{R}^2 \times (0,\infty),$$

where $a = a(x,t) \geq 0$ describes the flow rate of sand rolling downhill, that is, in the direction $-Du$. Suppose lastly that $\operatorname{spt} a \subseteq \{|Du| = 1\}$, so that the sand flows downhill only if the slope is one.

Show that the foregoing implies $f - u_t \in \partial I[u]$, for the convex function

$$I[u] := \begin{cases} 0 & \text{if } u \in L^2(\mathbb{R}^2),\ |Du| \leq 1 \text{ a.e.,} \\ \infty & \text{otherwise.} \end{cases}$$

13. U denotes a bounded, open subset of $\mathbb{R}^n$ with smooth boundary.

Assume u is a smooth solution of the gradient flow system (33) in §9.6.3, where L satisfies the uniform convexity condition (29). Show there exist constants $C, \gamma > 0$ such that

$$\int_U u_t^2(x,t)\,dx \leq Ce^{-\gamma t} \qquad (t > 0).$$

14. U denotes a bounded, open subset of $\mathbb{R}^n$ with smooth boundary.

Let u be a smooth solution of the nonlinear heat equation $u_t - \Delta\phi(u) = 0$ in $\mathbb{R}^n \times (0, \infty)$, where $\phi' > 0$. Assuming $|Du| > 0$, derive the formula

$$\frac{d}{dt}\int_{\mathbb{R}^n} |Du|\,dx = -\int_{\mathbb{R}^n} \frac{|D^2 v|^2}{|Dv|} - \frac{|D^2 v Dv|^2}{|Dv|^3}\,dx \leq 0$$

for $v := \phi(u)$.

10. Hamilton–Jacobi Equations

1. Assume u is a viscosity solution of $u_t + H(Du, x) = 0$ in $\mathbb{R}^n \times (0, \infty)$. Show that $\hat{u} := -u$ is a viscosity solution of $\hat{u}_t + \hat{H}(D\hat{u}, x) = 0$ in $\mathbb{R}^n \times (0, \infty)$, for $\hat{H}(p, x) := -H(-p, x)$.

2. Let $\{u^k\}_{k=1}^{\infty}$ be viscosity solutions of the Hamilton–Jacobi equations $u_t^k + H(Du^k, x) = 0$ in $\mathbb{R}^n \times (0, \infty)$ $(k = 1, \ldots)$, and suppose $u^k \to u$ uniformly. Assume as well that H is continuous. Show u is a viscosity solution of $u_t + H(Du, x) = 0$ in $\mathbb{R}^n \times (0, \infty)$.

Hence the uniform limits of viscosity solutions are viscosity solutions.

3. Suppose for each $\epsilon > 0$ that u^ϵ is a smooth solution of the parabolic equation

$$u_t^\epsilon + H(Du^\epsilon, x) - \epsilon \sum_{i,j=1}^{n} a^{ij} u^\epsilon_{x_i x_j} = 0$$

in $\mathbb{R}^n \times (0, \infty)$, where the smooth coefficients a^{ij} $(i, j = 1, \ldots, n)$ satisfy the uniform ellipticity condition from Chapter 6. Suppose also that H is continuous and that $u^\epsilon \to u$ uniformly as $\epsilon \to 0$.

Prove that u is a viscosity solution of $u_t + H(Du, x) = 0$. (This exercise shows that viscosity solutions do not depend upon the precise structure of the parabolic smoothing.)

4. Let u^i $(i = 1, 2)$ be viscosity solutions of

$$\begin{cases} u_t^i + H(Du^i, x) = 0 & \text{in } \mathbb{R}^n \times (0, \infty), \\ u^i = g^i & \text{on } \mathbb{R}^n \times \{t = 0\}. \end{cases}$$

Assume H satisfies condition (3) in §10.2. Prove the contraction property

$$\sup_{\mathbb{R}^n} |u^1(\cdot, t) - u^2(\cdot, t)| \leq \sup_{\mathbb{R}^n} |g^1 - g^2| \qquad (t \geq 0).$$

5.

(a) Show that $u(x) := 1 - |x|$ is a viscosity solution of

$$\begin{cases} |u'| = 1 & \text{in } (-1,1), \\ u(-1) = u(1) = 0. \end{cases} \tag{*}$$

This means that for each $v \in C^\infty(-1,1)$, if $u-v$ has a maximum (minimum) at a point $x^0 \in (-1,1)$, then $|v'(x^0)| \le 1$ (≥ 1).

(b) Show that $\tilde{u}(x) := |x| - 1$ is not a viscosity solution of (*).

(c) Show that $\tilde{u}$ is a viscosity solution of

$$\begin{cases} -|\tilde{u}'| = -1 & \text{in } (-1,1), \\ \tilde{u}(-1) = \tilde{u}(1) = 0. \end{cases} \tag{**}$$

(Hint: What is the meaning of a viscosity solution of (**)?)

(d) Why do problems (*), (**) have different viscosity solutions?

6. Let $U \subset \mathbb{R}^n$ be open, bounded. Set $u(x) := \text{dist}(x, \partial U)$ $(x \in U)$. Prove that u is Lipschitz continuous and that it is a viscosity solution of the eikonal equation $|Du| = 1$ in U.

This means that for each $v \in C^\infty(U)$, if $u - v$ has a maximum (minimum) at a point $x^0 \in U$, then $|Dv(x^0)| \leq 1$ (≥ 1).

7. Suppose an open set $U \subset \mathbb{R}^n$ is subdivided by a smooth hypersurface Γ into the subregions V^+ and V^-. Let ν denote the unit normal to Γ, pointing into V^+. Assume that u is a viscosity solution of $H(Du) = 0$ in U and that u is smooth in $\overline{V^+}$ and $\overline{V^-}$. Write u_ν^+ for the limit of $Du \cdot \nu$ along Γ from within V^+, and write u_ν^- for the limit from within V^-.

Prove that along Γ we have the inequalities

$$H(\lambda Du^- + (1-\lambda)Du^+) \geq 0 \quad \text{if } u_\nu^- \leq u_\nu^+$$

and

$$H(\lambda Du^- + (1-\lambda)Du^+) \leq 0 \quad \text{if } u_\nu^+ \leq u_\nu^-$$

for each $0 \leq \lambda \leq 1$, where $Du^\pm$ denote the gradients along Γ from $V^\pm$.

8. A surface described by the graph of $u : \mathbb{R}^2 \to \mathbb{R}$ is illuminated by parallel light rays from the vertical e_3 direction. We assume the surface has constant albedo and in addition is Lambertian, meaning that incoming right rays are scattered equally in all directions. Then the intensity $i = i(x)$ of the reflected light above the point $x \in \mathbb{R}^2$ is given by the formula $i = e_3 \cdot \nu$, where ν is the upward pointing unit normal to the surface.

Show that u solves a PDE of the form $|Du| = n$ for a given function $n = n(x)$, computed in terms of the intensity i. (Finding the surface by solving this PDE for u is the shape from shading problem.)

9. A yacht starts at the point $(x_1, 0)$ on the positive x_1-axis and sails to the right at speed $b_1 > 0$. Another yacht is initially at the point $(0, x_2)$ along the positive x_2-axis and starts in pursuit, sailing always towards the first yacht at speed $b_2 > b_1$.

Find the PDE solved by $u(x_1, x_2) :=$ time it takes the second yacht to intercept the first.

(Think of this as a dynamic programming problem, but with no controls.)

10. (Infinite horizon control problem) Assume $\mathbf{f}$ and r satisfy the conditions given in §10.3. Given a point $x \in \mathbb{R}^n$ and a control belonging to $\mathcal{A} := \{\boldsymbol{\alpha} : [0,\infty) \to A \mid \boldsymbol{\alpha}(\cdot) \text{ is measurable}\}$, let $\mathbf{x}(\cdot)$ be the unique solution of the ODE

$$\begin{cases} \dot{\mathbf{x}}(s) = \mathbf{f}(\mathbf{x}(s), \boldsymbol{\alpha}(s)) \quad (s > 0), \\ \mathbf{x}(0) = x. \end{cases}$$

Fix $\lambda > 0$ and define the discounted cost

$$C_x[\boldsymbol{\alpha}(\cdot)] := \int_0^\infty e^{-\lambda s} r(\mathbf{x}(s), \boldsymbol{\alpha}(s))\, ds.$$

Define the value function $u(x) := \inf_{\boldsymbol{\alpha}(\cdot)\in\mathcal{A}} C_x[\boldsymbol{\alpha}(\cdot)]$.

(a) Show that u is bounded and that if $\lambda > \mathrm{Lip}[\mathbf{f}]$, then u is Lipschitz continuous.

(b) Show that if $0 < \lambda \le \mathrm{Lip}[\mathbf{f}]$, then u is Hölder continuous for some exponent $0 < \alpha < 1$.

11. (Continuation) Prove that the value function u is a viscosity solution of the PDE

$$\lambda u - \min_{a \in A}\{\mathbf{f}(x,a) \cdot Du + r(x,a)\} = 0 \quad \text{in } \mathbb{R}^n.$$

(This means that if v is smooth and $u - v$ has a local maximum at a point x^0, then

$$\lambda u - \min_{a \in A}\{\mathbf{f}(x,a) \cdot Dv + r(x,a)\} \leq 0$$

at x^0, and that the opposite inequality holds if $u - v$ has a local minimum at x^0.)

12. Remember from §A.1 that if $R, S \in \mathbb{S}^n$, we write $R \geq S$ if $R - S$ is nonnegative definite. A function $F : \mathbb{S}^n \times \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$, $F = F(R, p, x)$, is elliptic provided $R \geq S$ implies $F(R, p, x) \leq F(S, p, x)$. Here $\mathbb{S}^n$ denotes the space of real, $n \times n$ symmetric matrices.

(a) Show that $F(R) = -\operatorname{tr} R$ is elliptic.

(b) More generally, show that if $A \in \mathbb{S}^n$ and $A \geq 0$, then $F(R) = -A : R = -\operatorname{tr}(AR^T)$ is elliptic.

(c) Show that if for each $k = 1, \ldots, m$, F_k is elliptic, then so are $\max_k F_k$ and $\min_k F_k$.

13. Let F be continuous and elliptic. We say that a function $u \in C(U)$ is a viscosity solution of the fully nonlinear elliptic PDE

$$F(D^2u, Du, x) = 0 \quad \text{in } U, \tag{*}$$

provided for each $v \in C^\infty(U)$, (i) if $u - v$ has a local maximum at a point $x^0 \in U$, then $F(D^2v(x^0), Dv(x^0), x^0) \leq 0$ and (ii) if $u - v$ has a local minimum at a point $x^0 \in U$, then $F(D^2v(x^0), Dv(x^0), x^0) \geq 0$.

Show that if u is a C^2 solution of (*), then u is a viscosity solution.

14. Assume that u_k is a viscosity solution of the elliptic PDE $F(D^2u_k, Du_k, x) = 0$ in U for $k = 1, \ldots$. Suppose $u_k \to u$ uniformly and show u is a viscosity solution of $F(D^2u, Du, x) = 0$ in U.

11. Systems of Conservation Laws

1. Show that this generalization of the p-system is a strictly hyperbolic system if $p', q' > 0$:

$$\begin{cases} u_t^1 - q(u^2)_x = 0, \\ u_t^2 - p(u^1)_x = 0. \end{cases}$$

2.

(a) Verify that the shallow water equations (Example 3 in §11.1) form a strictly hyperbolic system, provided $h > 0$.

(b) Show that for a smooth solution $(h, q) = (h, vh)$, with $h > 0$, the shallow water equations can be recast into this alternate conservation law form:

$$\begin{cases} h_t + (vh)_x = 0, \\ v_t + (\frac{v^2}{2} + h)_x = 0. \end{cases} \tag{*}$$

Check that this is a strictly hyperbolic system.

3. Define for $z \in \mathbb{R}$, $z \neq 0$, the matrix function

$$\mathbf{B}(z) := e^{-1/z^2} \begin{pmatrix} \cos(2/z) & \sin(2/z) \\ \sin(2/z) & -\cos(2/z) \end{pmatrix},$$

and set $\mathbf{B}(0) = 0$. Show that $\mathbf{B}$ is C^∞ and has real eigenvalues, but we cannot find unit-length right eigenvectors $\{\mathbf{r}^1(z), \mathbf{r}^2(z)\}$ depending continuously on z near 0. What happens to the eigenspaces as $z \to 0$?

4. (Rarefaction curves) Show that for the shallow water wave equations in the form (*) from Problem 2 the rarefaction curves R_1, R_2 in the (h, v)-plane are given by the formulas $2\sqrt{h} \pm v = c$, where c is a constant.

5. (Shock curves) For the shallow water wave equations in the variables (h, q), find a formula describing the shock set $S(z^0)$ in the (h, q)-plane, where $z^0 = (h^0, q^0)$ and $h^0 > 0$. For the particular case that $q^0 = 0$, show that the shock set is given by the expression

$$q = \pm \frac{(h - h^0)h}{\sqrt{2}} \left(\frac{1}{h} + \frac{1}{h^0} \right)^{1/2}.$$

6. (Continuation) Construct an entropy solution $\mathbf{u} = (h, q)$ of the shallow water equations, corresponding to the Riemann initial data $u_l = (h_l, 0)$ for $x < 0$ and $u_r = (h_r, 0)$ for $x > 0$, where $h_l > h_r > 0$.

Your answer should consist of a rarefaction wave connecting u_l to an intermediate state $u_i = (h_i, q_i) = (h_i, v_i h_i)$ and a shock wave connecting u_i to u_r. To do this, transfer the formula from Problem 5 to the (h, v)-plane, and show that there exists a point of intersection (h_i, v_i) of a rarefaction curve through $(h_l, 0)$ and a shock curve through $(h_r, 0)$.

7. Confirm that the functions w^1, w^2 computed for the barotropic gas dynamics in §11.3.1 are indeed Riemann invariants.

8. Suppose that Φ is an entropy for the shallow water equations in the form (*) from Problem 2. Prove $\frac{\partial^2 \Phi}{\partial v^2} = h \frac{\partial^2 \Phi}{\partial h^2}$.

9. Show that $\Phi = \rho v^2/2 + P(\rho)$ is an entropy for the barotropic Euler equations (from §11.3.1), provided $P''(\rho) = p'(\rho)/\rho$, $\rho > 0$. Confirm that Φ is convex in the proper variables. What is the corresponding entropy flux Ψ?

10. (Gradient flux function) Suppose that $\mathbf{F} = D\phi$, where $\phi : \mathbb{R}^m \to \mathbb{R}$. Show that $\Phi := \frac{1}{2}|z|^2$, $\Psi := z \cdot D\phi(z) - \phi(z)$ form an entropy/entropy-flux pair for the system of conservation laws $\mathbf{u}_t + \mathbf{F}(\mathbf{u})_x = 0$.

11. (Antigradient flux function)

(a) Assume that $m = 2$ and $\mathbf{F}$ has the "antigradient" form $\mathbf{F} = (\phi_{z_2}, \phi_{z_1})$ for a convex function $\phi : \mathbb{R}^2 \to \mathbb{R}$. Confirm that $\Phi := \phi$, $\Psi := \phi_{z_1}\phi_{z_2}$ are an entropy/entropy-flux pair for the system $\mathbf{u}_t + \mathbf{F}(\mathbf{u})_x = 0$.

(b) Find an entropy/entropy-flux pair for the generalization of the p-system given in Problem 1.

12. Formulate what it means for $\Phi, \boldsymbol{\Psi}$ to be an entropy/entropy-flux pair for the general system of conservation laws

$$\mathbf{u}_t + \operatorname{div} \mathbf{F}(\mathbf{u}) = 0 \quad \text{in } \mathbb{R}^n \times (0, \infty).$$

13. Maxwell's equations for nonlinear dielectrics read in part

$$\begin{cases} \mathbf{D}_t = \operatorname{curl} \mathbf{H}, \\ \mathbf{B}_t = -\operatorname{curl} \mathbf{E} \end{cases}$$

in $\mathbb{R}^3 \times (0, \infty)$, where $\mathbf{E}$ denotes the electric field; $\mathbf{H}$ the magnetic field; $\mathbf{D}$ the electric displacement; and $\mathbf{B}$ the magnetic induction. (We suppose the current $\mathbf{J}$ vanishes.) We introduce the constitutive relations

$$\mathbf{H} = D_B \eta(\mathbf{B}, \mathbf{D}), \qquad \mathbf{E} = D_D \eta(\mathbf{B}, \mathbf{D}),$$

determined by a given convex electromagnetic field energy $\eta = \eta(B, D)$. Show that $\Phi := \eta$, $\mathbf{\Psi} := D_D \eta \times D_B \eta$ are an entropy/entropy-flux pair.

14. (Nonconvex flux function) Assume that u is an entropy solution of the scalar conservation law $u_t + F(u)_x = 0$ and that, as in §3.4.1, u is smooth on either side of a curve $\{x = s(t)\}$.

(a) Prove that along this curve the left- and right-hand limits of u satisfy

$$F(\lambda u_l + (1-\lambda)u_r) \geq \lambda F(u_l) + (1-\lambda)F(u_r) \quad \text{if } u_l \leq u_r$$

and

$$F(\lambda u_l + (1-\lambda)u_r) \leq \lambda F(u_l) + (1-\lambda)F(u_r) \quad \text{if } u_r \leq u_l$$

for each $0 \leq \lambda \leq 1$. These inequalities are called condition E. Draw pictures illustrating the geometric meaning of these inequalities. (Compare with Problem 7 in Chapter 10.)

(b) What does condition E imply if F is uniformly convex?

15. (Explicit solutions for nonconvex flux) Compute the unique entropy solution of the Riemann problem

$$\begin{cases} u_t + (u^3 - u)_x = 0 & \text{in } \mathbb{R} \times (0, \infty), \\ u = g & \text{on } \mathbb{R} \times \{t = 0\}, \end{cases} \tag{*}$$

for $g(x) = -1$ if $x < 0$, $g(x) = 1$ if $x > 0$.

(Hint: The solution involves a rarefaction wave with a shock along its left edge. Make sure your solution verifies condition E, introduced in the previous problem.)

16. (Continuation) Discuss the structure of the entropy solution of the conservation law (*) from Problem 15, for

$$g(x) = \begin{cases} -1 + \lambda & \text{if } x < -1, \\ -1 - \lambda x & \text{if } -1 < x < 0, \\ 1 & \text{if } x > 0, \end{cases}$$

when $\lambda > 0$ is small. How does this perturbation affect the behavior of the shock in the solution found in the previous problem?

17. Assume u is a smooth solution with compact support in space of the scalar conservation law in several variables

$$\begin{cases} u_t + \operatorname{div} \mathbf{F}(u) = 0 & \text{in } \mathbb{R}^n \times (0, \infty), \\ u = g & \text{on } \mathbb{R}^n \times \{t = 0\}, \end{cases}$$

where $\mathbf{F} = (F_1, \dots, F_n)$. Show that if $\hat{u}$ is another smooth solution with initial data $\hat{g}$, we have the L^1-contraction estimate

$$\|u(\cdot, t) - \hat{u}(\cdot, t)\|_{L^1(\mathbb{R}^n)} \leq \|g - \hat{g}\|_{L^1(\mathbb{R}^n)}$$

for each time $t \geq 0$.

(Hints: Show that $w := u - \hat{u}$ satisfies $w_t + \operatorname{div}(\mathbf{b}w) = 0$ for an appropriate vector field $\mathbf{b} = \mathbf{b}(x, t)$. Let $T > 0$ and introduce the solution v of the adjoint problem $v_t + \mathbf{b} \cdot Dv = 0$ in $\mathbb{R}^n \times [0, T]$, with the terminal condition $v(\cdot, T) = \operatorname{sgn}(w(\cdot, T))$.)

12. Nonlinear Wave Equations

1. All given functions are assumed smooth, unless otherwise stated.

Assume u has compact support in space and solves the quasilinear wave equation

$$u_{tt} - \sum_{i=1}^{n} (L_{p_i}(Du))_{x_i} = 0 \quad \text{in } \mathbb{R}^n \times (0, \infty).$$

Determine the appropriate energy $E(t)$ and show $\dot{E} \equiv 0$.

2. All given functions are assumed smooth, unless otherwise stated.

Let u solve the Klein–Gordon equation

$$\begin{cases} u_{tt} - \Delta u + m^2 u = 0 & \text{in } \mathbb{R}^n \times (0, \infty), \\ u = g, \ u_t = h & \text{on } \mathbb{R}^n \times \{t = 0\}. \end{cases} \tag{*}$$

(a) Show that the energy

$$E(t) := \frac{1}{2} \int_{\mathbb{R}^n} u_t^2 + |Du|^2 + m^2 u^2 \, dx \qquad (t \geq 0)$$

is constant in time.

(b) Modify the proof in §4.3.1 showing asymptotic equipartition of energy for the wave equation to prove that $\lim_{t \to \infty} \int_{\mathbb{R}^n} |Du|^2 + m^2 u^2 \, dx = E(0)$.

3. All given functions are assumed smooth, unless otherwise stated.

Suppose u solves the initial value problem (*) from Problem 2 for the Klein–Gordon equation. Write $\bar{x} = (x, x_{n+1})$ for $x \in \mathbb{R}^n$ and define $\bar{u}(\bar{x}, t) := u(x, t)\cos(mx_{n+1})$.

(a) Show that $\bar{u}$ solves the wave equation $\Box\bar{u} = 0$ in $\mathbb{R}^{n+1} \times (0, \infty)$.

(b) Derive a formula for the solution of the initial-value problem for the Klein–Gordon equation when $n = 1$. (This is a variant of the method of descent, introduced in §2.4.1.)

4. All given functions are assumed smooth, unless otherwise stated.

Assume u solves $u_{tt} - \Delta u + du_t = 0$ in $\mathbb{R}^n \times (0, \infty)$, which for $d > 0$ is a damped wave equation. Find a simple exponential term that, when multiplied by u, gives a solution v of $v_{tt} - \Delta v + cv = 0$ for a constant $c < 0$. (This is the opposite of the sign for the Klein–Gordon equation.)

5. All given functions are assumed smooth, unless otherwise stated.

Check that for each given $y \in \mathbb{R}^n$, $y \neq 0$, the function $u = e^{i(x\cdot y - \sigma t)}$ solves the Klein–Gordon equation $u_{tt} - \Delta u + m^2 u = 0$ provided $\sigma = (|y|^2 + m^2)^{1/2}$. The phase velocity of this plane wave solution is $\sigma/|y| > 1$. Why does this not contradict the assertions in §12.1 that the speed of propagation for solutions is less than or equal to one?

6. All given functions are assumed smooth, unless otherwise stated.

Suppose

$$\begin{cases} u_{tt} - \Delta u = 0 & \text{in } \mathbb{R}^3 \times (0, \infty), \\ u = g,\ u_t = h & \text{on } \mathbb{R}^3 \times \{t = 0\}, \end{cases}$$

where g, h have compact support. Show there exists a constant C such that $|u(x,t)| \leq C/t$ ($x \in \mathbb{R}^3$, $t > 0$).

(Hint: Use the representation formula for the solution from §2.4.1.)

7. All given functions are assumed smooth, unless otherwise stated.

Let u solve

$$\begin{cases} u_{tt} - \Delta u = 0 & \text{in } \mathbb{R}^2 \times (0, \infty), \\ u = g, \ u_t = h & \text{on } \mathbb{R}^2 \times \{t = 0\}, \end{cases}$$

where g, h have compact support. Show that $|u(x,t)| \leq C/t^{1/2}$ ($x \in \mathbb{R}^2$, $t > 0$) for some constant C.

8. All given functions are assumed smooth, unless otherwise stated.

Suppose u^ε solves the linear wave equation in $n = 2$ space dimensions, with the initial conditions $u_t^\varepsilon = h \equiv 0$ and

$$u^\varepsilon = g_\varepsilon := \begin{cases} e^{-\varepsilon \frac{(r-2)^2}{(r-1)(3-r)}} & \text{if } 1 < r < 3, \\ 0 & \text{otherwise,} \end{cases}$$

where $r = |x|$. Show that although $|g_\varepsilon| \leq 1$, we have $\max_{\mathbb{R}^2 \times [0,4]} |u^\varepsilon| \to \infty$ as $\varepsilon \to 0$.

(Hint: Use (26) in §2.4.1 to compute $u^\varepsilon(0, t)$ for $t > 3$.)

9. All given functions are assumed smooth, unless otherwise stated.

(Kelvin transform for wave equation) The hyperbolic Kelvin transform $\mathcal{K}u = \bar{u}$ of a function $u : \mathbb{R}^n \times \mathbb{R} \to \mathbb{R}$ is

$$\begin{aligned}\bar{u}(x,t) &:= u(\bar{x},\bar{t})\big||\bar{x}|^2 - \bar{t}^2\big|^{\frac{n-1}{2}} \\ &= u\left(\frac{x}{|x|^2 - t^2}, \frac{t}{|x|^2 - t^2}\right) \frac{1}{\big||x|^2 - t^2\big|^{\frac{n-1}{2}}},\end{aligned}$$

provided $|x|^2 \neq t^2$, where $\bar{x} = \frac{x}{|x|^2 - t^2}$, $\bar{t} = \frac{t}{|x|^2 - t^2}$.

Show that if $\Box u = 0$, then $\Box \bar{u} = 0$. (Compare with Problem 11 in Chapter 2.)

10. All given functions are assumed smooth, unless otherwise stated.

Assume that u and v solve the system

$$\begin{cases} (u-v)_t = 2a\sin(\frac{u+v}{2}), \\ (u+v)_x = \frac{2}{a}\sin(\frac{u-v}{2}), \end{cases} \tag{*}$$

where $a \neq 0$. Show that both $w := u$ and $w := v$ solve the sine-Gordon equation in the form $w_{xt} = \sin w$.

Why is this equivalent to the PDE $\Box w = \sin w$?

11. All given functions are assumed smooth, unless otherwise stated.

(Continuation) Alternatively, given a solution v of the sine-Gordon equation, we can try to solve the system (*) to build a second solution u. This procedure is called a Bäcklund transformation.

Start with the trivial solution $v \equiv 0$, and use the Bäcklund transformation to compute for each choice of the parameter a another solution u. (Hint: First show u must have the form $f(at + x/a)$. Also show that $(\tan(u/4))_t = a \tan(u/4)$.)

12. All given functions are assumed smooth, unless otherwise stated.

Prove the Sobolev-type inequality (6) in §12.2.2.

(Hints: If $u \in H^k$ and β is a multiindex with $|\beta| \leq k$, we have the estimate $\|D^\beta u\|_{L^p} \leq C\|u\|_{H^k}$, where p satisfies (a) $p = \infty$ if $\frac{1}{2} - \frac{k}{n} + \frac{|\beta|}{n} < 0$; (b) $2 \leq p < \infty$ if $\frac{1}{2} - \frac{k}{n} + \frac{|\beta|}{n} = 0$; and (c) $\frac{1}{p} = \frac{1}{2} - \frac{k-|\beta|}{n}$ if $\frac{1}{2} - \frac{k}{n} + \frac{|\beta|}{n} > 0$.

Assume that the multiindices $|\beta_1|, \ldots, |\beta_r|$ satisfy (a) above, $|\beta_{r+1}|, \ldots, |\beta_s|$ satisfy (b), and $|\beta_{s+1}|, \ldots, |\beta_m|$ satisfy (c). Estimate $\|D^{\beta_1} u_1 \cdots D^{\beta_m} u_m\|_{L^2}^2$.)

13. All given functions are assumed smooth, unless otherwise stated.

A smooth function $\mathbf{u} : \mathbb{R}^n \times [0, \infty) \to \mathbb{R}^m$, $\mathbf{u} = (u^1, \dots, u^m)$, is called a wave map into the unit sphere $S^{m-1} = \partial B(0, 1) \subset \mathbb{R}^m$ provided that everywhere in $\mathbb{R}^n \times [0, \infty)$, $|\mathbf{u}| = 1$ and $\mathbf{u}_{tt} - \Delta \mathbf{u}$ is perpendicular to S^{m-1} at $\mathbf{u}$.

Show that therefore $\mathbf{u}$ solves the system of PDE $\mathbf{u}_{tt} - \Delta \mathbf{u} = (|D\mathbf{u}|^2 - |\mathbf{u}_t|^2)\mathbf{u}$.

14. All given functions are assumed smooth, unless otherwise stated.

Prove that if $\mathbf{u}$ is a wave map into the unit sphere, with compact support in space, we have conservation of energy:

$$\frac{d}{dt}\int_{\mathbb{R}^n} |\mathbf{u}_t|^2 + |D\mathbf{u}|^2\, dx = 0.$$

15. All given functions are assumed smooth, unless otherwise stated.

(Small energy for $p = 5$) Adapt the proof of Theorem 3 in §12.3.3 to show the existence of a smooth solution of

$$\begin{cases} u_{tt} - \Delta u + u^5 = 0 & \text{in } \mathbb{R}^3 \times (0, \infty), \\ u = g, \ u_t = h & \text{on } \mathbb{R}^3 \times \{t = 0\}, \end{cases}$$

provided the energy $E(0)$ is sufficiently small.

(Hint: Modify estimate (24) in §12.3.3 by introducing the L^6-norm of g.)

16. All given functions are assumed smooth, unless otherwise stated.

Let u be a complex-valued solution of the nonlinear Schrödinger equation

$$iu_t + \Delta u = f(|u|^2)u \quad \text{in } \mathbb{R}^n \times (0, \infty). \tag{*}$$

Here $f : \mathbb{R} \to \mathbb{R}$. Demonstrate that if $\xi \in \mathbb{R}^n$, then $w(x,t) := e^{\frac{i}{4}(2\xi \cdot x - |\xi|^2 t)} u(x - \xi t, t)$ also solves the NLS equation. This shows the Galilean invariance of solutions.

17. All given functions are assumed smooth, unless otherwise stated.

Assume u solves the nonlinear Schrödinger equation (*) from Problem 16 and decays rapidly, along with its derivatives, as $|x| \to \infty$. Derive these identities:

$$\frac{d}{dt}\int_{\mathbb{R}^n} |u|^2\,dx = 0 \qquad \text{(conservation of mass)},$$

$$\frac{d}{dt}\int_{\mathbb{R}^n} |Du|^2 + F(|u|^2)\,dx = 0 \qquad \text{(conservation of energy)},$$

where $F' = f$, and

$$\frac{d}{dt}\int_{\mathbb{R}^n} \frac{\bar{u}Du - uD\bar{u}}{2i}\,dx = 0 \qquad \text{(conservation of momentum)}.$$

Remember that $|u|^2 = u\bar{u}$.

18. All given functions are assumed smooth, unless otherwise stated.

(Continuation)

(a) Under the hypotheses of the previous problem, derive the identity

$$\frac{d^2}{dt^2}\int_{\mathbb{R}^n}|x|^2|u|^2\,dx = 8\int_{\mathbb{R}^n}|Du|^2\,dx + 4n\int_{\mathbb{R}^n}f(|u|^2)|u|^2 - F(|u|^2)\,dx.$$

(b) Use (a) to show that there does not exist a solution of the cubic-NLS equation $iu_t+\Delta u+|u|^2u = 0$ in $\mathbb{R}^n \times (0,\infty)$ existing for all times $t \geq 0$, if

$$E(0) = \int_{\mathbb{R}^n}|Du(x,0)|^2 - \frac{|u(x,0)|^4}{2}\,dx < 0$$

and $n \geq 2$. (R. Glassey, *J. Math. Physics* 18 (1977), 1794–1797)